

ActewAGL Distribution

May 2011

ActewAGL

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Contents

List of tables	ii
Overview	iii
1. Introduction	1
1.1 Purpose and scope of the document	1
1.2 Background	1
1.3 Review of Final Distribution Determination	2
1.4 Structure of this document	2
2. The structure and basis of ActewAGL Distribution's network tariffs	5
2.1 Network tariff structure	5
2.2 Pricing strategy	12
2.3 Consistency with the transitional Rules pricing principles	14
2.4 The price setting process	19
3. Network tariffs for 2011/12	23
3.1 The revenue cap for standard control services	23
3.2 Distribution use of system charges	24
3.3 Miscellaneous services charges	28
3.4 Transmission use of system charges	30
3.5 Network use of system charges	33
3.6 Changes to network tariffs	35
4. Charges for metering services	41
4.1 The structure and basis of ActewAGL Distribution's metering charges	41
4.2 Metering services charges for 2011/12	41
4.3 Changes in metering charges	42
5. Indicative customer impacts	45
5.1 Changes in network and metering charges	45
5.2 Estimated impacts on average customer electricity bills	46
Attachment 1: Compliance with transitional Rules requirements	47

List of tables

Table 2.1	Network tariff structure – residential.....	7
Table 2.2	Network tariff structure - commercial low voltage	9
Table 2.3	Network tariff structure - high voltage.....	11
Table 2.4	Avoidable and stand alone costs 2011/12 (\$'000)	16
Table 2.5	Long run marginal costs per network level (\$/kVA pa).....	17
Table 2.6	Long run marginal costs by tariff class (\$/kVA pa)	18
Table 2.7	Allocating service costs to tariff charging parameters.....	21
Table 3.1	Calculation of the Maximum Allowable Average Revenue 2011/12	24
Table 3.2	Calculation of the revenue cap for DUOS 2011/12	24
Table 3.3	Distribution use of system charges 2011/12	26
Table 3.4	Weighted average DUOS revenue by tariff class.....	28
Table 3.5	Miscellaneous fees and charges 2011/12	29
Table 3.6	TUOS overs and unders account.....	30
Table 3.7	Transmission use of system charges 2011/12.....	32
Table 3.8	Network use of system charges 2011/12	34
Table 3.9	Changes in network charges.....	37
Table 3.10	Changes in miscellaneous services charges	39
Table 4.1	Alternative control services revenue cap	41
Table 4.2	Metering charges 2011/12.....	42
Table 4.3	Changes in metering charges	43
Table 5.1	Network and metering charges 2011/12	45

Overview

ActewAGL Distribution offers customers a range of network tariff options across three tariff classes—residential, commercial low voltage and high voltage. Customers are able to choose the tariff that best suits their needs, subject to some eligibility requirements as set out in this document.

As noted by the Australian Energy Regulator (AER) in the *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision* (the Final Distribution Determination), over the course of the 2009-14 regulatory period, ActewAGL Distribution will significantly increase investment on its network, which will result in higher prices for electricity consumers in the ACT. The AER also noted that increases in network charges are being driven by the introduction of the ACT feed-in tariff scheme. 2011/12 network tariffs are influenced also by increases in charges for transmission services, which must be passed on to customers in accordance with Chapter 11 of the *National Electricity Amendment Rules* (the transitional *Rules*).

The distribution use of system (DUOS) charges for 2011/12 are 0.29 cents per kWh, or 5.7 per cent, higher on average than those for 2010/11. Transmission use of system (TUOS) charges, levied on ActewAGL Distribution by TransGrid, are 0.21 cents per kWh, or 16.6 per cent, higher on average than the charges for 2010/11. The network use of system (NUOS) charges (comprising both DUOS and TUOS) for 2011/12 are, as a result, on average 0.50 c/kWh, or 7.8 per cent, higher than the average charges for 2010/11. Charges for miscellaneous services for 2011/12 are around 3.1 per cent higher than the 2010/11 charges.

Charges for metering services are subject to a separate regulatory cap. Metering charges for 2011/12 are 0.3 per cent lower on average than the charges for 2010/11.

ActewAGL Distribution estimates that the 2011/12 increases in network and metering charges will raise the electricity bill for an average residential customer consuming 7000 kWh on the residential basic network charge by \$0.77 per week (including GST)—a real increase of 4.2 per cent (7.2 per cent nominal). For a commercial customer consuming 30 MWh per annum on the general network charge, the network and metering price increases would raise their electricity bill by \$4.10 per week (including GST)—a 3.8 per cent real increase (6.7 per cent nominal).

1. Introduction

1.1 Purpose and scope of the document

ActewAGL Distribution has prepared this document in accordance with the requirements in Chapter 11 of the *National Electricity Amendment Rules* (the transitional *Rules*). It provides the required information on the tariffs and charges to apply to ActewAGL Distribution's regulated distribution services from 1 July 2011 to 30 June 2012. A checklist of the transitional *Rules* requirements and where they are met in this document is provided in Attachment 1.

The document relates to tariffs and charges to apply to ActewAGL Distribution's standard control services and alternative control services, as defined in clause 6.2.3C of the transitional *Rules*. ActewAGL Distribution's standard control services comprise distribution network use of system services and miscellaneous distribution services. ActewAGL Distribution's alternative control services comprise the provision and servicing of all manually read meters for consumers with loads less than 160 MWh per annum.

In accordance with the transitional *Rules*, separate regulatory control mechanisms apply to standard control and alternative control services. Therefore, separate price schedules must be determined. The combined standard control (network) and alternative control (metering) prices are also provided in this document.

The retail component of the delivered price of electricity to consumers accounts for around 55 per cent of the total price. The retail provision of electricity is fully open to competition, although there remains a regulated retail tariff offer, set through a separate retail regulatory process, for customers consuming less than 100 MWh per year.

1.2 Background

The Australian Energy Regulator (the AER) is responsible for the economic regulation of distribution services provided by ActewAGL Distribution. The AER has determined the maximum allowed average revenue (MAAR) for ActewAGL Distribution's standard control services and a revenue cap for alternative control (metering) services for the 2009-14 regulatory period.¹

This document should be read in conjunction with the AER's Final Distribution Determination² and ActewAGL Distribution's submissions and proposals (all published on the AER's website), as they set out in detail the basis of the costs that are reflected in the prices in this document. The AER summarises the outcomes of the regulatory review as follows:

¹ AER, *Australian Capital Territory distribution determination 2009-10 to 2013-14: Final Decision*, 28 April 2009

² Subject to the part of the Final Distribution Determination that is subject to a pending application to the Federal Court: see section 1.3

Over the course of the next regulatory control period, ActewAGL will significantly increase investment on its network, which will result in higher prices for electricity consumers in the ACT. Higher prices will be largely driven by significant investment in four major capital projects. These projects include construction of two new zone substations, which are the first to be built in the ACT since 1994, augmentation of a third substation and construction of new assets to improve the security of electricity supply to the ACT. In addition, ActewAGL has already undertaken significant capital works to reinforce and replace a large number of unsafe poles, and these costs will now flow through to consumers. While consumers within the ACT will face higher charges as a result of this investment, they will also benefit from a more reliable and secure network.³

The AER also notes that, in addition to higher capital investment, increases in network charges are being driven by the introduction of the feed-in tariff scheme, which imposes costs on ActewAGL Distribution.⁴

1.3 Review of Final Distribution Determination

ActewAGL Distribution submitted its 2011/12 pricing proposal to the AER on 29 April 2011. The network charges included in the pricing proposal were calculated using an X factor consistent with ActewAGL Distribution's 23 September 2010 application to the Federal Court of Australia for review of a discrete part of the AER's Final Distribution Determination. A decision by the Court in this application is pending.

The AER wrote to ActewAGL Distribution by letter dated 6 May 2011 stating that it had determined that ActewAGL Distribution's pricing proposal was deficient and that ActewAGL Distribution is required to correct the deficiencies by setting prices consistent with the X factor in the AER's Final Distribution Determination. This document is provided in response to that letter, but it is provided under protest and subject to a number of other qualifications contained in a covering letter to AER dated 20 May 2011. Further, the pricing proposal dated 29 April 2011 is not withdrawn.

1.4 Structure of this document

ActewAGL Distribution's tariff structure for standard control services is set out in chapter 2. The chapter includes details on the components and rationale for each tariff, an outline of ActewAGL Distribution's pricing strategy and how it relates to the pricing principles in the transitional *Rules*, and an explanation of the price setting process—that is, the process of moving from the regulated maximum allowed average revenue and a set of high level pricing principles to a full schedule of tariffs and charges for 2011/12.

The network tariffs and charges for ActewAGL Distribution's standard control services for 2011/12 are presented in chapter 3. The chapter includes discussion of the changes relative to 2010/11.

³ AER 2009, *Australian Capital Territory Distribution Determination 2010/11 to 2013/14: Final Decision*, April, p. viii

⁴ AER 2009, *Australian Capital Territory Distribution Determination 2010/11 to 2013/14: Final Decision*, April, p. viii

The structure and basis of ActewAGL Distribution's charges for alternative control (metering) services, the charges for 2011/12 and the changes relative to 2010/11 are presented and explained in chapter 4.

Indicative estimates of the likely impacts of the price changes on average customer electricity bills are provided in chapter 5.

2. The structure and basis of ActewAGL Distribution's network tariffs

The transitional *Rules* (clause 6.18.2) require a description of the *tariff classes*⁵ and tariffs that are to apply in 2011/12. For each tariff within a tariff class, the *charging parameters*⁶ and the elements of service to which they relate must also be set out in the pricing proposal.

2.1 Network tariff structure

ActewAGL Distribution's offers network tariffs in three tariff classes:

- Residential;
- Commercial low voltage (LV); and
- High voltage (HV).

The transitional *Rules* stipulate that tariff classes must be constituted with regard to the need to group customers together on an economically efficient basis and the need to avoid unnecessary transactions costs (clause 6.18.3(d)). ActewAGL Distribution meets this requirement by grouping customers according to type of connection (residential or commercial), and connection voltage (LV or HV). Customers within each class have similar load and connection characteristics. The relevant costs for each class can then be identified and reflected in the tariffs for each class.

Within each of these three tariff classes, ActewAGL Distribution has developed a suite of network tariffs that effectively meet the diverse needs of its customer base, encourage efficient use of the network and signal the costs of future network expansion. Residential customers are offered a choice of four network tariff options plus two controlled load off-peak options and an embedded renewable generation tariff option. Commercial LV customers are offered four main tariff options. Commercial customers on the general network charge also have access to the controlled load off-peak tariff options and the embedded renewable generation tariff option on a similar basis to customers in the residential class. Commercial HV customers are offered four tariff options. Customers are able to choose the option which best suits their needs, subject to the eligibility criteria set out in Tables 2.1 to 2.3 below.

The network tariffs comprise different combinations of the following *charging parameters*:

- Network access charges—these apply per customer for residential consumers and per connection point for commercial customers. They involve a fixed daily charge and do not vary with electricity consumption or capacity;
- Energy charges—these apply to each unit of electricity consumed. The c/kWh rate may vary with the level of consumption (with higher rates applying above certain thresholds) or with the time-of-use (with lower rates applying at off-peak periods);

⁵ A tariff class is defined in chapter 10 of the *National Electricity Rules* as “a class of customers for one or more direct control services who are subject to a particular tariff or particular tariffs”.

⁶ Charging parameters are defined as “the constituent elements of a tariff” in chapter 10 of the *National Electricity Rules*.

- Maximum demand charges—these apply per connection point for some commercial tariffs. They involve a charge per unit of maximum demand (in c/kVA/day). The maximum demand is the highest demand calculated over a 30-minute interval during the billing period;
- Capacity charges—these apply on the same basis as maximum demand charges, but are for the maximum demand recorded over the previous 12 months.

Network access charges relate to the connection services provided to customers. They are associated with the costs of the provision and use of connection assets for each tariff class, as well as customer related costs such as fault call centre costs.

Energy charges relate to the distribution services provided to customers. They are linked to the costs of the provision and use of distribution assets (other than connection assets), and also recover most of the common services costs. Higher energy rates at peak periods reflect higher costs. Higher energy rates beyond 330 kWh per day for the general network charge encourage larger consumers with a good load factor to move to demand or time-of-use network charges.

Maximum demand and capacity charges relate to the capacity provided to meet maximum demand and encourage consumers to manage their load on the network.

The allocation of costs to charging parameters is discussed further in section 2.4 below.

The tariffs and charging parameters for each tariff class are shown in the following tables (2.1 to 2.3). The tables include an explanation of the purpose of each tariff and the customers to which each tariff may apply.

2.1.1 Network tariffs for residential customers

ActewAGL Distribution's residential network tariff structure is shown in Table 2.1. The *Residential time-of-use (TOU)*, *Residential 5000* and *Residential with heat pump* tariffs are refinements of the *Residential basic* tariff to reflect customer load profiles. The *Residential TOU* tariff provides an opportunity and an incentive for customers with the necessary metering capability to respond to price signals at different times of the day and so reduce their electricity bill in line with the costs they impose on the network. The *Residential 5000* and *Residential with heat pump* tariffs involve a higher connection charge and an inclining block structure with a higher energy charge (cents per kWh) applying above certain thresholds. These tariffs more accurately tailor costs to the load profile of these customers. The off-peak tariff options can be used in conjunction with the *Residential basic* and the *Residential TOU* network charges.

Table 2.1 Network tariff structure – residential

Tariff	Charging parameters	Explanation
Residential basic network	Network access charge (c/day/customer) Energy charge (c/kWh)	The residential basic network tariff is available to installations at private dwellings, excluding serviced apartments, but including: • Living quarters for members and staff of religious orders; • Living quarters on farms; • Charitable homes; • Retirement villages; • Residential sections of nursing homes and hospitals; • Churches, buildings or premises which are primarily used for public worship; and • Approved caravan sites. The energy charge varies neither with the level of consumption nor the time of day. However, customers on this tariff are also eligible for the off-peak tariffs.
Residential time-of-use (TOU) network ⁷	Network access charge (c/day/customer) Energy at max times, ie 7am to 9am and 5pm to 8pm every day (c/kWh) Energy at mid times, ie 9am to 5pm and 8pm to 10pm every day (c/kWh) Energy at economy times, ie all other times (c/kWh)	This tariff is available to residential customers (as defined above) with a meter able to be read as a time-of-use meter and to electric vehicle recharge facilities on residential premises. The energy charges relate to the supply of network services at various times. Higher rates apply at max or peak times to encourage users to shift their load to off-peak periods. Customers on this tariff are also eligible for the controlled load off-peak tariffs.
Residential 5000 network	Network access charge (c/day/customer) Energy for the first 60 kWh/day (c/kWh) Energy above 60 kWh/day (c/kWh)	This tariff is designed for residential customers who have large continuous (rather than time controlled) loads, such as electric hot water systems, and consume over 5,000 kWh per annum. The energy charges relate to the supply of network services above and below certain volume thresholds. An inclining block structure applies (ie higher energy rates for the second block of energy). The lower energy rate is limited to consumption up to 60 kWh per day, reflecting a typical domestic usage profile. This is sufficient to cover the energy requirements of many residential consumers.
Residential with heat pump	Network access charge (c/day/customer) Energy for the first 165 kWh/day (c/kWh) Energy above 165 kWh (c/kWh)	This tariff is only available to residential customers with a reverse cycle air conditioner. An inclining block structure applies (ie higher energy rates for the second block of energy). The lower energy rate is set to recover the incremental cost of energy load on the network as a demand management tool to lower winter peak loads and improve utilisation of the network in summer and so improve overall network utilisation.
Off-peak (1) night network	Energy at controlled times, ie between 10 pm and 7 am (c/kWh)	The off-peak (1) night charge is available only to customers utilising a controlled load element, and taking all other energy at residential basic network, residential time-of-use or general network rates. It is not available where the customer's meter is read as an interval meter. The off-peak (1) night charge is applicable to permanent heat (or cold) storage; electric vehicle recharge; and CNG vehicle gas compression installations. The design and rating must be acceptable to ActewAGL. The installation must use most energy during the controlled times

⁷ All times for metering are Eastern Standard Time.

Tariff	Charging parameters	Explanation
		but may be boosted at the principal charge, or charges, at other times. The off-peak (1) night network energy charge relates to supply of network services at controlled times, for 6 to 8 hours per day between the hours of 10 pm and 7 am.
Off-peak (3) day and night network	Energy at controlled times, ie between 10 pm and 7 am and 9 am and 5 pm (c/kWh)	Available only to customers utilising a controlled load element, and taking all other energy at residential basic network, residential time-off-use or general network rates. This charge is not available where the customer's meter is read as an interval meter. This charge is applicable to permanent heat (or cold) storage installations. The design and rating must be acceptable to ActewAGL. The off-peak (3) day and night network energy rate applies to power supplied for up to 13 hours per day between 10 pm and 7 am and again between 9 am and 5 pm.
Renewable generation	Energy charges (c/kWh)	This tariff applies to customers with grid connected solar or wind energy generation systems. Different arrangements apply to customers participating in the ACT feed-in tariff scheme, in accordance with the <i>Electricity Feed-in (Renewable Energy Premium) Act 2008</i> (ACT).

2.1.2 Network tariffs for low voltage commercial customers

ActewAGL Distribution sets different tariffs for commercial low voltage (LV) and commercial high voltage (HV) customers recognising the different costs associated with supplying each group. Within the commercial LV tariff class a range of tariff options has been developed to meet the diverse needs of commercial customers and to accommodate their differing load profiles and ability to respond to price signals. Of the four main options offered to commercial LV customers, all but the *General network* tariff involve time-of-use charges. The *General network* tariff does however involve an inclining block tariff structure with higher energy charges (c/kWh) applying above certain thresholds. Also, the off-peak (controlled load) tariffs are available to customers on the *General network* tariff.

Two of the commercial LV options involve capacity and/or maximum demand charges, in conjunction with time-of-use charges. Customers able to improve their load factor⁸ have an incentive to choose a tariff with a demand or capacity charge, and reduce their energy bills. Customers on the *General network* and *General time-of-use network* charges can move to the demand charges and lower their network costs if they have a sufficiently large load (for the cost savings to offset the higher cost of interval metering) and a reasonable load profile.

⁸ The load factor is the actual energy consumed relative to the energy the customer would have consumed if it had consumed at its maximum rate of consumption for the full period.

Table 2.2 Network tariff structure - commercial low voltage

Tariff	Charging parameters	Explanation
General network	Network access charge (c/day/customer) Energy for the first 330 kWh/day (c/kWh) Energy above 330 kWh/day (c/kWh)	This tariff is available to all customers. The tariff is most suitable for small commercial consumers operating in regular business hours or larger customers with poorer load factors (peaky loads). This tariff may be used in conjunction with the off-peak tariffs.
General TOU network	Network access charge (c/day/customer) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is particularly suitable for small commercial customers with discretionary or relatively large off-peak loads such as bakers, freezer installations and irrigators. The energy charges relate to supply of network services at different times.
LV TOU kVA demand network	Network access charge (c/day/connection point). Maximum demand (in billing period) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for customers with an average or stable commercial load. The maximum demand charge is designed to encourage consumers to manage their demand upon the network. The energy charges relate to supply of energy at different times, with lower rates in off-peak times reflecting the availability of capacity and encouraging consumers to shift their load from peak to off-peak times to utilise the available capacity.
LV TOU capacity network	Network access charge (c/day/connection point) Maximum demand (in billing period) (c/kVA/day) Capacity (max demand in last year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is intended for low voltage customers with embedded generation but it may be an attractive option for other low voltage customers with seasonally stable loads. The tariff provides an incentive for customers with embedded generation to manage their output and their down-times (eg for servicing) so as to minimise their demand on the network.
Streetlighting	Network access charge (c/day/customer) Energy at any time (c/kWh)	This tariff applies to the night-time lighting of streets and public ways and places.
Small unmetered loads	Network access charge (c/day/customer) Energy at any time (c/kWh)	This tariff applies to eligible installations as determined by ActewAGL, including: • telephone boxes • telecommunication devices • other, as determined by the National Metrology Coordinator. Energy charges are calculated based on the assessed rating of the load and the charge period.
Internal (Obsolete)	Network access charge (c/day/customer) Energy at any time (c/kWh)	This tariff applies only to ActewAGL Distribution sites and fully recovers the economic costs of supply. It is set at the rate of the first step in the general network tariff.

* Business times are between 7 am and 5 pm Eastern Standard Time on weekdays. Evening times are between 5 pm and 10 am Eastern Standard Time on weekdays. Off-peak times are all other times.

2.1.3 Network tariffs for high voltage customers

To qualify for the high voltage demand network charges, consumers must take their energy at high voltage (nominal voltage not less than 11 kV) and make a capital contribution towards their connection assets and transformers. High voltage consumers have the option of owning and operating their own high voltage assets. Some customers have aggregated their load, incorporating part of ActewAGL Distribution's low voltage network to become a high voltage customer. A separate high voltage network charge is available for such customers.

Customers taking their energy at high voltage also have the option of selecting the network tariffs available to low voltage consumers. For example, a high voltage customer with a poor load factor could select the *General* or the *General time-of-use* network charge.

Table 2.3 Network tariff structure - high voltage

Tariff (code)	Charging parameters	Explanation
HV TOU Demand Network (111)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for large customers taking supply at high voltage with a low voltage network owned and maintained by ActewAGL. The energy charges relate to supply of network services at different times, with lower rates in off-peak times encouraging customers to increase their utilisation of the network in off-peak periods. The demand charge is applied to the maximum demand in the billing period while the capacity charge is applied to the maximum demand in the previous 12 months. The capacity charge encourages consumers to monitor and manage their peak demand over the year while the demand charge continues to encourage consumers to manage their demand requirements each month.
HV TOU Demand Network – Customer HV (112)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for large customers taking supply at high voltage with a low voltage network owned and maintained by ActewAGL, where the customer owns and is responsible for their high voltage assets (including transformers and switch gear).
HV TOU Demand Network – Customer LV (121)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This network charge is appropriate for large customers taking supply at high voltage where the customer owns and is fully responsible for their own low voltage network. The network access charge relates to the connection services provided to the customer including provision of the current transformer necessary to meter these large loads. The capacity charge encourages the consumer to monitor and manage their peak demand over the year while the demand charge continues to encourage consumers to manage their capacity requirements each month. The energy charges relate to supply of network services at different times, with lower rates in off-peak times reflecting the relatively low costs of off-peak supply, and thereby providing incentives for customers to switch their utilisation of the network to off-peak periods.
HV TOU Demand Network – Customer HV and LV (122)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This network charge is appropriate for large customers taking supply at high voltage where the customer owns and is fully responsible for their own low voltage network and where the customer owns and is responsible for their high voltage assets (including transformers and switch gear).

* Business times are between 7 am and 5 pm Eastern Standard Time on weekdays. Evening times are between 5 pm and 10 am Eastern Standard Time on weekdays. Off-peak times are all other times.

2.1.4 Miscellaneous charges and fees

In addition to the network tariffs set out above, ActewAGL Distribution offers a range of miscellaneous services with charges for each service. These services are part of ActewAGL Distribution's standard control services, as set out in Appendix D of the Final Distribution Determination. The charges are regulated through the maximum allowed average revenue cap applying to standard control services. The structure of each charge depends on the type of service. Some services are charged on a per visit basis, others per installation or per test. The charges for miscellaneous services are set on a cost reflective basis. For example, separate rates apply for temporary connections depending on whether they relate to an overhead, standard underground or free-standing underground connection, as these will involve different costs. Standard control miscellaneous services fees and charges are discussed further in chapter 3.

ActewAGL Distribution also provides some miscellaneous services as part of its alternative control (metering) services. These include meter tests, installation of interval meters at a customer's request, and special meter reads. These are covered in chapter 4.

2.2 Pricing strategy

ActewAGL Distribution has developed and refined its network tariff structure over time, guided by its pricing strategy. The strategy involves:

- Setting prices to signal to customers the economic costs of providing distribution services;
- Providing customers with a choice of flexible and innovative tariffs to best meet their needs;
- Providing incentives and opportunities for demand management;
- Ensuring that tariffs are set to recover costs in a way that encourages efficient use of the network and signals to customers the cost of network expansion; and,
- Offering customers a clear and simple tariff structure, noting the need to take account of the ability of different customer groups to respond to price signals and the need to keep transactions costs low.

ActewAGL Distribution's pricing strategy has in recent years accommodated the development of some innovative tariffs and significant customer responses. For example, in line with the strategies of setting cost reflective prices and providing opportunities and incentives for demand management, ActewAGL Distribution has gradually introduced several time-of-use charging options for both commercial and residential customers. More than 50 per cent of the total load in the ACT is now subject to time-of-use or controlled load (off-peak) charges. For the non-residential sector, 80 per cent of the load is on time-of-use or controlled load tariffs.

The introduction of maximum demand charges in several commercial tariff options has further strengthened price signals to customers, provided incentives to use the network more efficiently and resulted in significant customer response. The maximum demand charges signal to customers the relatively high cost of providing capacity to meet demand and provide incentives to customers to improve both their load factor (that is, spread their load more

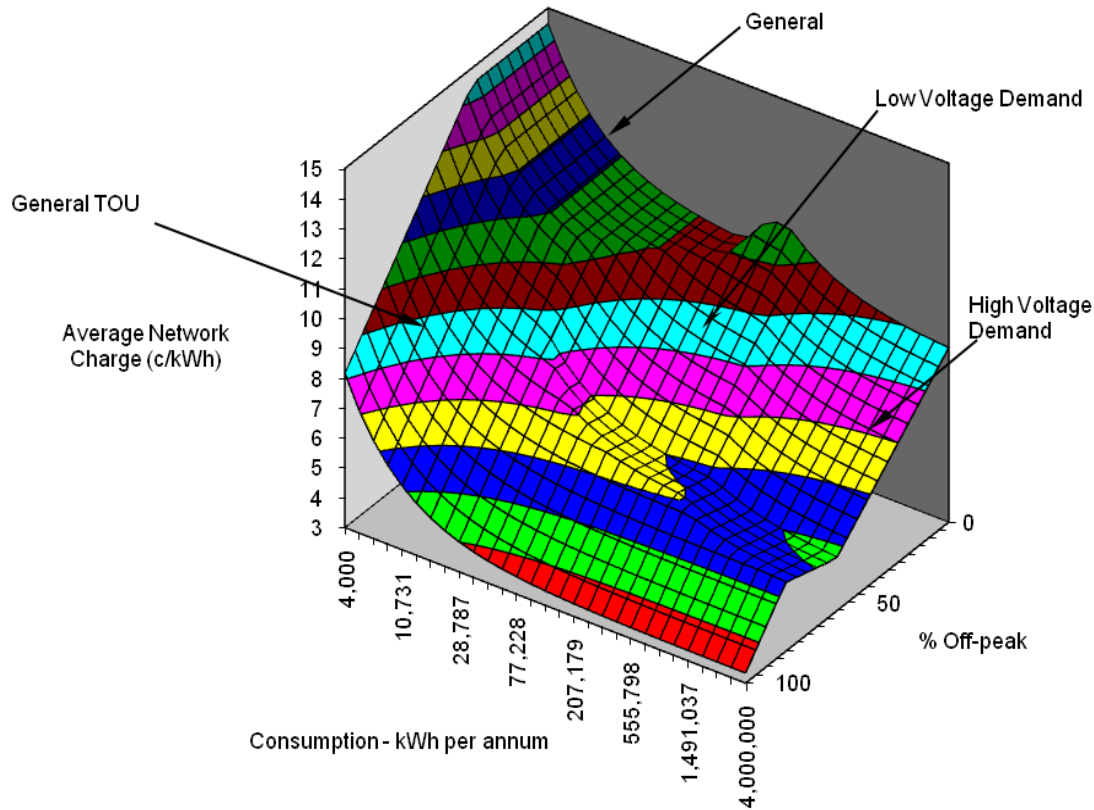
evenly) and power factor (which allows the existing network to deliver more energy). Between 1999/00 and 2009/10, customers on the *Low voltage demand* network tariff improved their load factor and therefore their utilisation of the network by 10.6 per cent, increasing the average energy consumed relative to the average of their monthly maximum demand from 40.1 per cent to 44.4 per cent. Over the same period, high voltage customers increased their load factor, and therefore their utilisation of the network, from 54.2 per cent to 58.9 per cent, an improvement of 8.6 per cent.

These price signals have been effective demand management tools and have allowed ActewAGL Distribution to keep network augmentation costs to a minimum. For example, until the current regulatory period, no new zone substations had been built since 1994.

The options available for commercial customers and the incentives created are represented in Figure 2.1 which shows the main network tariffs available to commercial customers, including time-of-use and demand tariffs. Sophisticated modelling ensures that the tariffs are synchronised and that the costs of supplying particular loads are appropriately priced. ActewAGL Distribution is able to offer a suite of commercial tariffs and be indifferent to customers' choices between them.

The graph plots the average minimum network charges over a range of consumption levels and a range of consumption profiles from all *peak time* (business and evening) to all *off-peak* for commercial customers. It shows how the *General network* charge suits small commercial customers with relatively large peak time loads, while those with high off-peak loads can be rewarded by selecting the *General time-of-use network* tariff. The graph also demonstrates how the low voltage and high voltage demand tariffs align with the non-demand tariffs. Larger customers with good load profiles are able to choose a demand based tariff option and reduce overall network supply costs and their average network charges.

Figure 2.1 Average network charges for commercial customers



2.3 Consistency with the transitional Rules pricing principles

The transitional *Rules* include pricing principles to be applied when setting tariffs (clause 6.18.5). These principles recognise the complexities which arise in electricity network pricing because of the nature of electricity distribution costs—in particular the large fixed costs associated with providing capacity to meet peak demand, the high incidence of shared or joint costs, and costs which vary according to the time of day and season and voltage of supply. There is no unique set of efficient and equitable prices, but rather a range within which prices should fall, taking account of the need to raise sufficient revenue to cover costs, the need to signal to customers the costs of expanding capacity to meet future demand, and the need to take account of the scope for customers to respond to price signals, bearing in mind the administrative and technical costs of doing so.

Clause 6.18.5(a) of the transitional *Rules* requires that for each tariff class the revenue must lie between an upper bound of the *stand alone costs* and a lower bound of *avoidable cost*. The stand alone cost for any group of customers is the cost that would be incurred if only that customer group were supplied. Any costs that would otherwise be shared with other customer groups would have to be fully attributed to the stand alone customers. The stand alone cost is

effectively the cost of replicating or bypassing the infrastructure.⁹ The avoidable cost for any group of customers is the cost that would be avoided if the group of customers were removed from the network.

The purpose of the *between stand alone and avoidable cost* test is to ensure that there are no cross subsidies between tariff classes. Compliance with the test discourages inefficient bypass of the network (which may occur if prices are above the stand alone cost).¹⁰ If any tariff class, or group of customers, is paying less than the avoidable cost lower bound—the costs that arise directly from its use of the network—then it is receiving a subsidy. If any tariff class is paying more than the stand alone cost upper bound—the cost if it alone were supplied—then it is subsidising other users. If the revenues from each group lie between the bounds of avoidable cost and stand alone cost then each group is making a contribution to covering the joint or shared costs of supply, while no group is receiving or paying an economic subsidy.

Clause 6.18.5(b) of the transitional *Rules* says that each tariff and charging parameter must take account of the long run marginal cost of the service (or element of the service) to which it relates, while also taking account of transactions costs and whether customers are able or likely to respond to price signals. The purpose of the long run marginal cost requirement is to ensure that prices signal to customers the forward-looking costs of meeting additional demand or the savings from reduced demand.

Clause 6.18.5(c) of the transitional *Rules* says that if, as a result of the operation of clause 6.18.5(b), expected revenue may not be covered, tariffs must be adjusted in a way that results in minimum distortion to efficient patterns of consumption.

ActewAGL Distribution's pricing strategy is consistent with each of the pricing principles in clause 6.18.5 of the transitional *Rules*. As far as possible, costs are allocated to the customer group or groups which are directly responsible for the costs. For example, the costs associated with providing low voltage connection assets are directly attributed to low voltage customers, but not high voltage customers. As a result, the revenue from each tariff class recovers at least the avoidable cost (or directly attributed cost) associated with providing the service to that class. Where additional costs must be recovered, each class pays a share, but none pays more than its stand alone cost. In this way, the requirement of providing subsidy-free prices is met. This is demonstrated in section 2.3.1 below, where revenue from each tariff class is shown to lie between the relevant bounds of avoidable and stand alone costs.

The principle of signalling to users the long run marginal cost is also taken into account in ActewAGL Distribution's approach to pricing. The long run marginal cost is the additional cost of meeting additional future demand. It includes the capital costs associated with any increment in capacity required to meet the additional demand, as well as the additional operating and maintenance costs. At times when demand is at or near full network capacity, prices should signal the costs of expanding capacity rather than the short run marginal costs which do not include capital related costs.

⁹ NERA 2006, *Distribution Pricing Rule Framework*, Report prepared for the Network Policy Working Group of the Standing Committee of Officials (SCO) of the Ministerial Council on Energy (MCE), December, p. 22

¹⁰ NERA 2006, *Distribution Pricing Rule Framework*, December, p. 22

ActewAGL Distribution's capacity and maximum demand charges provide signals to commercial customers about the cost of capacity expansion and, therefore, encourage these users to manage their maximum demands. The *General time-of-use* tariff does not incorporate a demand charge, but it does signal to customers the relatively high cost of providing capacity at peak times.

While residential tariffs do not have capacity or maximum demand charges, a time-of-use tariff is available and off-peak tariff options can be combined with other tariffs (including the *Residential basic* tariff). The higher energy rates at peak periods signal to customers the cost of expanding capacity and, therefore, encourage consumers to shift their load to off-peak periods. The residential time-of-use tariff is now the default tariff for new residential consumers.

When setting tariffs to ensure that all costs are recovered, ActewAGL Distribution also aims to minimise the distortion to efficient patterns of consumption. This influences the balance between the network access charges (which do not vary with the level of consumption) and the energy charges (which vary with the level of consumption and therefore more directly influence consumption patterns). For example, ActewAGL Distribution seeks to minimise the fixed charges in the *General time-of-use* and the *Low voltage demand* tariffs to ensure that small customers are not inefficiently constrained from adopting the tariffs with time-of-use and demand price signals. Also, for residential customers, ActewAGL Distribution has kept the fixed charge (including metering charge) for the *Residential time-of-use* charge the same as that for the *Basic residential* charge to ensure there is not an inefficient barrier to customers seeking to move to a tariff that provides time-of-use price signals.

The practical application of these strategies and principles in the price setting process is described in section 2.4 below.

2.3.1 Calculating avoidable, stand alone and long run marginal costs

ActewAGL Distribution engaged economic consultants NERA in 2009 to advise on the application of the pricing principles and to provide estimates of avoidable costs, stand alone costs and long run marginal costs. NERA undertook a detailed analysis of the available methods and developed and implemented a recommended approach to calculating avoidable, stand alone and long run marginal costs

ActewAGL Distribution has used NERA's estimates of avoidable cost for 2009/10 for each tariff class escalated in line with the increase in revenue for each customer class. These costs have been used to calculate the stand alone costs for each tariff class using NERA's methodology. The results for avoidable and stand alone costs are shown in Table 2.4. The table also shows that average 2011/12 DUOS revenue for each tariff class lies within the lower bound of avoidable cost and the upper bound of stand alone costs. The tariffs therefore comply with the requirement in clause 6.18.5(a) of the transitional *Rules*.

Table 2.4 Avoidable and stand alone costs 2011/12 (\$'000)

Tariff class	Avoidable cost ('000)	DUOS charges ('000)	Stand alone cost ('000)
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Residential	\$9,261	\$60,256	\$154,943
Commercial low voltage	\$1,143	\$81,979	\$146,825
High voltage	\$58	\$13,909	\$145,740
Total		\$156,144	

NERA's analysis of long run marginal costs highlights a range of practical and theoretical issues that need to be addressed in the calculation and interpretation of long run marginal cost, particularly where the distribution network may not be in a 'steady-state'.¹¹ ActewAGL Distribution considers that these matters should be carefully examined in the context of the AER's development of guidelines for the calculation of stand alone, avoidable and long run marginal costs (in accordance with clause 6.2.8 of the transitional *Rules*).

Following a detailed review of available methods and methods used in practice, NERA developed for ActewAGL Distribution a long run marginal cost model based on the average incremental cost (AIC) approach. Recognising that the long run marginal cost of network expansion varies depending on the network type (that is, voltage level), NERA estimates the long run marginal cost for each network type then translates this into a cost estimate for each customer class on the basis of the contribution of each class to demand at times of network peak demand.¹²

NERA's estimates of long run marginal cost by network level, on a \$/kVA per year basis, are shown in Table 2.5. The estimates are expressed separately for each network level, with the low voltage estimate broken down to reflect the long run marginal cost of network assets used by low voltage commercial customers, those used by low voltage residential customers and shared low voltage assets.

Table 2.5 Long run marginal costs per network level (\$/kVA pa)

<i>Network level</i>	<i>Long run marginal cost \$/kVA pa</i>
Low voltage residential	239.57
Low voltage commercial	29.76
Low voltage shared	4.74
High voltage	113.91

NERA's estimates of long run marginal cost for each tariff class are shown in Table 2.6. These long run marginal cost estimates represent the costs of meeting an additional kVA of demand for each of the three tariff classes:

- For customer located in the HV network, the long run marginal cost reflects the cost of augmenting the high voltage network, in order to meet that additional demand;

¹¹ For example, while the Turvey approach is generally considered to have the stronger theoretical basis, the Average Incremental Cost (AIC) method is more widely used as it is easier to apply. However, the AIC method has important shortcomings in that it is based on average capital costs associated with an increment in demand, not marginal, and the results are also sensitive to the demand forecasts used and the time frame for the analysis.

¹² NERA 2010, *Analysis of ActewAGL's Electricity Distribution Services Costs, A report for ActewAGL*, commercial-in-confidence, April, p. 11

- For a low voltage commercial customer, the long run marginal cost reflects the cost of augmenting those areas of the low voltage network used exclusively by those customers, plus the costs of augmenting those areas of the low voltage network which are shared between low voltage commercial and residential customers and plus the costs of augmenting the high voltage network;
- Similarly, the long run marginal cost of meeting an increase in residential customer demand reflects the costs of augmenting those areas of the network used exclusively by residential customers plus those parts of the low voltage network which are shared between low voltage commercial and residential customers and plus the costs of augmenting the high voltage network.

As a result of applying this method of allocating costs, the long run marginal cost estimates for the low voltage customer classes are above the long run marginal cost estimates for the high voltage customer class.

Table 2.6 Long run marginal costs by tariff class (\$/kVA pa)

Tariff class	Long run marginal cost \$/kVA pa
Residential	367.06
Commercial low voltage	157.25
High voltage	113.91

ActewAGL Distribution notes that the long run marginal cost estimates are significantly influenced by a number of data limitations as well as features of the ACT demand and expenditure profiles in the current regulatory period. These features reflect ActewAGL Distribution's stage in the investment cycle—that the distribution network is not in a steady-state—and the lumpy nature of network investment. Care must therefore be taken when interpreting the results and drawing implications for tariff design.

For example, some of the expected future increase in electricity demand is associated with extensions of ActewAGL Distribution's existing distribution network to new residential subdivisions. NERA explains that, conceptually, the capital expenditure associated with extending the current network to meet additional demand in new areas can be distinguished from the capital expenditure associated with meeting increments in demand on the existing network. It is the latter which should be reflected in the estimate of long run marginal cost, as this provides the relevant behavioural signal to consumers. However in practice it is difficult to isolate incremental demand in existing network areas from demand forecasts for ActewAGL Distribution's overall network. As a result NERA calculated the long run marginal cost on the basis of total additional demand and total capital augmentation, including extension of the network to new areas.¹³ The unique circumstances in the current regulatory period, involving a significant increase in ActewAGL Distribution's forecast total capital augmentation costs, together with very low forecast growth rates for residential demand, result in proportionately

¹³ NERA 2010, *Analysis of ActewAGL's Electricity Distribution Services Costs, A report for ActewAGL*, commercial-in-confidence, April, p. 20

higher estimates of long run marginal cost for residential customers. For these reasons, and consistent with the obligation to take long run marginal cost into account, the marginal cost estimate must be carefully interpreted and refined to reflect actual capacity availability in existing residential areas, otherwise it would inadvertently provide a signal to existing customers that there were capacity constraints when the existing residential network has sufficient capacity to meet demand.

2.4 The price setting process

The process of moving from the maximum allowed average revenue (MAAR) as set by the AER to a set of network tariffs involves the following steps:

1. Determine the maximum revenue to be recovered through distribution use of system (DUOS) charges. This requires some adjustments to the MAAR set by the AER in the final decision, as described in section 2.4.1 below.
2. Allocate the total revenue requirement to cost pools, taking account of the cost drivers (for example, whether the costs are demand related or customer related) and the type of assets involved (high voltage or low voltage). This cost allocation process is described in section 2.4.2 below.
3. Allocate the costs to tariff classes via a combination of network access charges, energy charges, demand charges and capacity charges. Costs are allocated according to the type of connection (residential, commercial, LV or HV) and the load profile. This process of setting the distribution use of system (DUOS) charges for each tariff class is described in section 2.4.3 below.
4. Allocate transmission use of system (TUOS) charges to tariff classes, to determine the total charges (DUOS + TUOS) to apply for each tariff class. The process of allocating TUOS charges is described in section 2.4.4 below.

2.4.1 Revenue to be recovered through DUOS charges

In accordance with the AER's Final Distribution Determination (chapter 4, section 4.5) and guideline for standard control services (appendix B, table 1), the maximum allowed average revenue for 2011/12 (in c/kWh) is converted to a total allowed revenue for 2011/12 by multiplying by 2010 throughput (in kWh). Approved cost pass-throughs (positive or negative) are then added to this allowed revenue. Miscellaneous services charges are deducted, since, although part of the MAAR for standard control services, they are set separately as described above. The resulting value is the total maximum amount to be recovered through DUOS charges.

The relevant values for each of these components and the calculation of the DOUS cap for 2011/12 are provided in chapter 3 of this document.

2.4.2 Allocation to cost pools

The total amount to be recovered in 2011/12 is allocated to three cost pools:

- Connection service costs
- Energy, demand and capacity service costs
- Common services costs

The main component of *connection service costs* is the cost of providing connection services to customers—that is, the cost associated with service mains (the wires that connect the customer's premises to the network).¹⁴ The cost pool also includes other costs that are related to the number of customers or connections, for example fault call centre costs.

The connection service costs are allocated across tariff classes using the proportion of connections in each tariff class relative to the total number of connections.

The main components of *energy, demand and capacity service costs* are the costs of reticulation and providing capacity. The cost pool includes the costs associated with subtransmission assets (transformers, 132 kV substation assets) HV assets (for example, high voltage distribution lines) and LV assets (LV substations and reticulation assets). The split between HV and LV costs is important as it ensures that LV and HV customers are each allocated the appropriate costs—for example, HV customers without any ActewAGL LV reticulation are not required to pay for low voltage mains reticulation. This cost pool includes some shared costs which are demand-related, or related to the size of the network (for example, training and apprenticeships costs), but which cannot be attributed to certain voltages.

The energy, demand and capacity service costs are allocated across tariff classes on a cost reflective basis using energy consumed at different parts of the load cycle (business, evening and off-peak times), based on the method of intercepts.

Common services costs are shared across all users on the basis of energy consumption.

2.4.3 Allocating costs to tariff charging parameters

The cost of connection services and other customer related costs are generally recovered through the network access charges. Energy delivery, demand and capacity service costs are recovered through the energy, maximum demand and capacity charging parameters for those commercial tariffs where these components apply. Costs for common services are recovered in energy charges.

The allocation from cost pools to charging parameters is summarised in Table 2.7.

¹⁴ Metering assets and costs are regulated separately under the alternative control services control mechanism.

Table 2.7 Allocating service costs to tariff charging parameters

	Cost pools		
	Connection service costs	Energy, demand and capacity service costs	Common service costs
Residential			
Network access charge	Residential connection service costs and other customer related costs are recovered through network access charges.	For the <i>Residential 5000</i> and <i>Residential with heat pump</i> network charges, some demand and capacity service costs are recovered via network access charges.	
Energy charges		Energy delivery, demand and capacity service costs for residential customers are generally recovered through residential energy charges.	Common service costs are recovered in energy charges.
Commercial LV			
Network access charge	Commercial connection service costs and other customer related costs are recovered through network access charges.		
Energy charges		Energy delivery service costs are recovered through the energy charge charges. Demand and capacity service costs are also recovered in energy charges where demand and capacity charging parameters are not applied.	Common service costs are recovered through the energy charges.
Capacity and maximum demand charges		Where demand and capacity charges apply, demand service costs and capacity service costs are recovered in demand and capacity charges.	
Commercial HV			
Network access charge	High voltage connection service costs and other customer related costs are recovered through network access charges		
Energy charges		Energy delivery costs are recovered through the energy charges.	Common service costs are recovered through the energy charges.
Capacity and maximum demand charges		Demand and capacity service costs are recovered through the demand and capacity charges.	

2.4.4 Allocating transmission use of system charges

In keeping with the strategy of setting prices to reflect the economic costs of providing network services, ActewAGL Distribution aims to preserve the economic signals present in the structure of TransGrid's transmission use of system (TUOS) charges when allocating the charges to tariff classes and components. However, approximately 70 per cent of TUOS charges to the ACT are now on a postage stamp basis—only 30 per cent are on a cost reflective basis.

TransGrid's transmission charges comprise a fixed charge, an energy charge (with general and common service components) and a monthly maximum demand charge. The transmission exit charge is about 1.0 per cent of the total and varies according to the size of the connection. ActewAGL Distribution recovers this component of the transmission charge in its network energy charges or its demand and capacity charges (where possible) thereby ensuring that it is allocated according to the customer's utilisation of the transmission network. The transmission energy charges are also recovered in the network energy charges. For commercial tariffs that include a maximum demand charge, the demand component of the transmission charge is recovered in the maximum demand charge. In all other cases TransGrid's maximum demand charge is recovered in the network energy charges. In this way, ActewAGL's prices preserve the economic signals present in the structure of the TUOS charges where appropriate.

3. Network tariffs for 2011/12

3.1 The revenue cap for standard control services

3.1.1 Base maximum allowable average revenue for standard control services

ActewAGL Distribution's standard control services prices are regulated using a maximum allowable average revenue (MAAR) cap. To calculate the maximum allowable average revenue for 2011/12, the following formula is used, as required by the AER's Final Distribution Determination:¹⁵

$$\text{Maximum allowable average revenue per kWh (MAAR}_t) = \text{MAAR}_{t-1} \times (1 + \text{CPI}_t) \times (1 - X_t)$$

where:

- MAAR_{t-1} is the maximum allowable average revenue per kWh in the previous year; and
- CPI means the Consumer Price Index, All Groups Index number for the weighted average of eight capital cities.

The CPI figure will be determined using the following formula:

$$\text{CPI}_t = \frac{\text{CPI}_{\text{March}_{t-2}} + \text{CPI}_{\text{June}_{t-2}} + \text{CPI}_{\text{Sept}_{t-1}} + \text{CPI}_{\text{Dec}_{t-1}}}{\text{CPI}_{\text{March}_{t-3}} + \text{CPI}_{\text{June}_{t-3}} + \text{CPI}_{\text{Sept}_{t-2}} + \text{CPI}_{\text{Dec}_{t-2}}} - 1$$

where:

- year t is the year for which tariffs are being set; and
- year t – 1 is the previous regulatory year.

The CPI adjustment figure for the year commencing 1 July 2011 is calculated as follows:

$$\text{CPI}_{(2011)} = \frac{\text{CPI}_{\text{March}_{2010}} + \text{CPI}_{\text{June}_{2010}} + \text{CPI}_{\text{Sept}_{2010}} + \text{CPI}_{\text{Dec}_{2010}}}{\text{CPI}_{\text{March}_{2009}} + \text{CPI}_{\text{June}_{2009}} + \text{CPI}_{\text{Sept}_{2009}} + \text{CPI}_{\text{Dec}_{2009}}} - 1$$

Therefore:

$$\text{CPI}_{2011} = \frac{171.0 + 172.1 + 173.3 + 174.0}{162.2 + 167.0 + 168.6 + 169.5} - 1$$

$$\text{CPI}_{2011} = \frac{690.4}{671.3} - 1$$

$$\text{CPI}_{2011} = 2.85\%$$

¹⁵ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 18

The maximum average revenue for 2010/11 calculated according to the AER's price determination for that year is \$0.05147 per kWh. This document incorporates an X factor of -4.00 per cent, consistent with the AER's Final Decision, Australian Capital Territory distribution determination 2009-10 to 2013-14. The CPI of 2.85 per cent is applied to the maximum allowed average revenue (MAAR) for 2010/11 to obtain the maximum allowed average revenue in 2011/12. The calculations of the MAAR are shown in Table 3.1.

Table 3.1 Calculation of the Maximum Allowable Average Revenue 2011/12

	<i>MAAR previous year</i>	<i>X Factor</i>	<i>Sum of CPI indices</i>	<i>CPI</i>	<i>MAAR</i>
2008/09			631.8		\$0.04092
2009/10	\$0.04092	-13.82%	659.3	4.35%	\$0.04860
2010/11	\$0.04860	-4.00%	671.3	1.82%	\$0.05147
2011/12	\$0.05147	-4.00%	690.4	2.85%	\$0.05505

Note that, while the CPI is shown as a percentage to 2 decimal places, the actual CPI figure applied the MAAR of the previous year is calculated based on the sum of the CPI indices for each year divided by the sum of the CPI indices for the previous year. Also, while the MAAR for 2008/09 is rounded to 5 decimal places, the MAAR figures for subsequent years are not rounded.

The maximum allowable average revenue cap is applied to the actual energy transported in the previous calendar year to establish a price cap for the following financial year. Therefore, the prices for 2011/12 are based upon energy transported in 2010. The actual energy transported in the 2010 calendar year was 2,899,994,192 kWh. When this is applied to the maximum allowable average revenue for 2011/12 of \$0.05505 per kWh, it results in the revenue cap for standard control services delivered in 2010 of \$159,639,065 which forms the regulated cap for 2011/12 prices.

3.1.2 Calculation of the revenue cap for DUOS

The calculation of the revenue to be recovered from 2011/12 distribution use of system (DUOS) charges is shown in Table 3.2.

Table 3.2 Calculation of the revenue cap for DUOS 2011/12

Maximum allowable average revenue 2011/12 (\$/kWh)*	A	\$0.05505
Energy sales ACT 2010 (kWh)	B	2,899,994,192
Standard control services revenue cap	$C = A \times B$	\$159,639,065
Approved cost pass-throughs/Feed-in tariff adjustment [#]		-\$2,171,932
Total standard control services revenue cap	$E = C + D$	\$157,467,133
Less miscellaneous fees and charges	F	\$1,323,058
Distribution use of system revenue cap	$G = E - F$	\$156,144,075

* The MAAR shown in this table has been rounded to 5 decimal places. The calculations have been made without rounding.

3.2 Distribution use of system charges

Distribution use of system (DUOS) prices for 2011/12 as shown in Table 3.3. These would have recovered \$156,144,065 on the actual customer, demand and energy quantities

recorded in the 2010 calendar year. The table shows the tariff classes that are to apply in 2011/12 and the tariffs for each tariff class. The charging parameters for each tariff are set out together with the service to which that charging parameter relates. All prices exclude GST unless otherwise stated.

To demonstrate compliance with the AER's control mechanism, ActewAGL Distribution is required to demonstrate that the sum of the standard control services revenue using the prices for the pricing year and the quantities for the previous calendar year divided by the quantity of energy in kWh transported over the previous calendar year is less than or equal to the MAAR for the pricing year.¹⁶

The sum of the distribution use of system charges, the feed-in tariff adjustment and miscellaneous fees is \$159,639,054. When this is divided by the energy transported in the ACT of 2,899,994,192 kWh, it results in an average price of \$0.05505 per kWh. As this average price is equal to the MAAR, the prices comply with the AER's Final Distribution Determination.

¹⁶ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 18

Table 3.3 Distribution use of system charges 2011/12

<i>Tariff & element of service</i>	<i>Units</i>	<i>2010 kWh/ cust no/ kVA</i>	<i>Charges 2011/12</i>	<i>Notional revenue 2011/12</i>
Residential tariffs				
010 Residential Basic Network				
Network access	c/day/customer	130,120	15.25	\$7,242,784
Energy at any time	c/kWh	926,040,359	4.933	\$45,681,571
015 Residential TOU Network				
Network access	c/day/customer	178	15.25	\$9,925
Energy at max times	c/kWh	830,207	7.461	\$61,942
Energy at mid times	c/kWh	953,105	3.694	\$35,208
Energy at economy times	c/kWh	1,496,274	2.600	\$38,903
020 Residential 5000 Network				
Network access	c/day/customer	8,352	36.75	\$1,120,300
Energy for the first 60 kWh per day	c/kWh	66,372,997	3.353	\$2,225,487
Energy above 60 kWh per day	c/kWh	2,376,217	4.933	\$117,219
030 Residential with Heat Pump Network				
Network access	c/day/customer	4,573	79.95	\$1,334,521
Energy for the first 165 kWh per day	c/kWh	82,003,467	1.863	\$1,527,725
Energy above 165 kWh per day	c/kWh	1,207,635	4.933	\$59,573
060 Off-Peak (1) Night Network				
Energy at controlled times	c/kWh	15,878,911	0.350	\$55,576
070 Off-Peak (3) Day & Night Network				
Energy at controlled times	c/kWh	89,804,626	0.830	\$745,378
Renewable Energy Generation				
Energy generated & consumed	c/kWh	465,590	0.000	\$0
Commercial low voltage tariffs				
040 General Network				
Network access	c/day/customer	12,319	30.71	\$1,380,805
Energy for the first 330 kWh per day	c/kWh	278,164,407	7.850	\$21,835,906
Energy above 330 kWh per day	c/kWh	26,542,647	10.749	\$2,853,069
135 Small Unmetered Loads Network				
Network access	c/day/customer	23	24.80	\$2,082
Energy at any time	c/kWh	1,352,100	9.331	\$126,164
141 Internal Network (Obsolete)				
Network access	c/day/customer	98	30.71	\$10,994
Energy at any time	c/kWh	706,316	7.850	\$55,446
080 Streetlighting Network				
Network access	c/day/customer	18	30.00	\$1,934
Energy for night time lighting of streets public ways & places	c/kWh	40,151,345	5.097	\$2,046,514
090 General TOU Network				
Network access	c/day/customer	303	30.71	\$33,915
Energy at business times	c/kWh	16,441,439	13.152	\$2,162,378
Energy at evening times	c/kWh	5,915,492	6.371	\$376,876
Energy at off-peak times	c/kWh	18,748,938	2.337	\$438,163
Low voltage time of use demand network				

<i>Tariff & element of service</i>	<i>Units</i>	<i>2010 kWh/ cust no/ kVA</i>	<i>Charges 2011/12</i>	<i>Notional revenue 2011/12</i>
101 LV TOU kVA Demand Network				
Network access	c/day/connection point	1,610	36.00	\$211,550
Maximum demand	c/kVA/day	232,185	34.81	\$29,500,647
Energy at business times	c/kWh	364,538,852	3.660	\$13,342,122
Energy at evening times	c/kWh	125,965,345	2.195	\$2,764,939
Energy at off-peak times	c/kWh	409,484,521	0.800	\$3,275,876
103 LV TOU Capacity Network				
Network access	c/day/connection point	20	36.00	\$2,619
Maximum demand (in billing period)	c/kVA/day	6,520	16.03	\$381,511
Capacity (maximum demand in last year)	c/kVA/day	7,297	16.03	\$426,935
Energy at business times	c/kWh	12,478,504	3.660	\$456,713
Energy at evening times	c/kWh	5,729,845	2.195	\$125,770
Energy at off-peak times	c/kWh	20,719,128	0.800	\$165,753
High voltage tariffs				
High voltage time of use demand network with ActewAGL low voltage network				
Network access	\$/day/connection point	1	13.00	\$4,745
Maximum demand (in billing period)	c/kVA/day	2,466	13.91	\$125,212
Capacity (maximum demand in last year)	c/kVA/day	3,039	13.91	\$154,307
Energy at business times	c/kWh	4,223,276	3.071	\$129,697
Energy at evening times	c/kWh	1,659,634	1.780	\$29,541
Energy at off-peak times	c/kWh	5,338,577	0.649	\$34,647
Network access	\$/day/connection point	0	13.00	\$0
Maximum demand (in billing period)	c/kVA/day	0	12.91	\$0
Capacity (maximum demand in last year)	c/kVA/day	0	12.91	\$0
Energy at business times	c/kWh	0	3.071	\$0
Energy at evening times	c/kWh	0	1.780	\$0
Energy at off-peak times	c/kWh	0	0.649	\$0
High voltage time of use demand network without ActewAGL low voltage network				
Network access	\$/day/connection point	21	13.00	\$100,451
Maximum demand (in billing period)	c/kVA/day	70,412	13.91	\$3,574,902
Capacity (maximum demand in last year)	c/kVA/day	86,838	13.91	\$4,408,898
Energy at business times	c/kWh	138,441,490	2.661	\$3,683,928
Energy at evening times	c/kWh	52,713,533	1.420	\$748,532
Energy at off-peak times	c/kWh	183,249,415	0.499	\$914,415
Network access	\$/day/connection point	0	13.00	\$0
Maximum demand (in billing period)	c/kVA/day	0	12.91	\$0
Capacity (maximum demand in last year)	c/kVA/day	0	12.91	\$0

<i>Tariff & element of service</i>	<i>Units</i>	<i>2010 kWh/ cust no/ kVA</i>	<i>Charges 2011/12</i>	<i>Notional revenue 2011/12</i>
Energy at business times	c/kWh	0	2.661	\$0
Energy at evening times	c/kWh	0	1.420	\$0
Energy at off-peak times	c/kWh	0	0.499	\$0
Total				\$156,144,065
Total Customers		157,635		
Total Energy Consumption		2,899,994,192		

Table 3.4 sets out for each tariff class related to standard control services the expected weighted average DUOS revenue¹⁷ for the relevant regulatory year and the current regulatory year, as required by clause 6.18.2(b)(4) of the transitional *Rules*.

Clause 6.18.6 of the transitional *Rules* applies a side constraint on the expected weighted average revenue to be raised from tariff classes. The side constraint is given by the formula $(1 + \text{CPI})(1 - X)(1 + 2\%)$. For 2011/12, the CPI is 2.85 per cent and the X factor is -4.00 per cent. Therefore, the side constraint is 9.10 per cent. As shown in Table 3.4, the price increase for each of the three tariff classes is within the side constraint.

Table 3.4 Weighted average DUOS revenue by tariff class

<i>Tariff class</i>	<i>2010/11 (c/kWh)</i>	<i>2011/12 (c/kWh)</i>	<i>Change %</i>
Residential	4.8	5.1	5.8%
Commercial Low Voltage	5.8	6.2	5.9%
High Voltage	3.5	3.6	3.8%
Average	5.1	5.4	5.7%

3.3 Miscellaneous services charges

As set out in the AER's guideline for standard control services, prices for miscellaneous services are regulated under the standard control services revenue cap. Like the DUOS charges, the charges for miscellaneous services for year "t" are checked for compliance with the revenue constraint for year "t" prices by multiplying the prices for the coming year by actual sales in the previous calendar year, for each service.¹⁸

Charges for miscellaneous services are shown in Table 3.5. These charges, when applied to units sold in 2010, would have recovered \$1,323,058.

¹⁷ This is the total DUOS revenue for each class divided by the total energy delivered to each class.

¹⁸ AER 2008, *Guideline on control mechanisms for direct control services for the ACT and NSW 2009 distribution determinations*, February, p. 5.

Table 3.5 Miscellaneous fees and charges 2011/12

Code	Description	Units	Units 2010	Prices 2011/12	Revenue with 2011/12 prices
For visit to re-energise or de-energise a site					
501	Business hours – re-energise a site	per visit	5,332	\$50.91	\$271,441
502	After hours – re-energise a site	per visit	1,756	\$109.09	\$191,562
503	Business hours – de-energise a site	per visit	3,297	\$44.55	\$146,881
505	De-energise a site for non-payment	per visit	1,053	\$84.55	\$89,014
508	Field visit only (De-energise a site for non-payment)	per visit	180	\$60.91	\$10,956
Temporary connections					
520	Overhead	per installation	105	\$360.91	\$37,784
521	Standard underground	per installation	14	\$523.64	\$7,416
522	Free-standing underground	per installation	6	\$635.45	\$4,000
Services					
530	Overhead: remove, reposition or disconnect service	per site visit	556	\$260.00	\$144,545
531	Underground: remove, reposition or disconnect service	per site visit	505	\$625.45	\$315,897
Upgrade service from single to three phase at customer's request where load does not justify three phase					
532	Overhead	per installation	3	\$335.45	\$930
533	Underground – service cable replacement not required	per installation	1	\$335.45	\$260
534	Underground – service cable replacement required	per installation	0	\$625.45	\$0
540	Rescheduled visit (applied where a revisit to a site is necessitated by obstructed access, non-compliance with the Service and Installation Rules or the client is not ready for the scheduled work.)	per visit	5	\$172.73	\$864
550	Issue of copies of electrical drawings	per sheet or electronic equivalent	300	\$13.45	\$4,039
560	De-energising wires (to allow safe approach, for example, for tree pruning, plant operation, oversize loads, construction activities)	per occurrence	234	\$417.27	\$97,469
570	Embedded generation - charges for small embedded generators (excluding renewable < 30KW)				
	for operation and maintenance of connection assets	per annum applied to the value of connection assets appreciated each year by CPI		2.00%	\$0
	for operation and maintenance of shared network assets (if applicable)	per annum applied to the value of shared assets allocated to the generator appreciated by CPI		2.00%	\$0
Total standard control miscellaneous services					\$1,323,058

3.4 Transmission use of system charges

The AER separately regulates transmission use of system (TUOS) charges. TUOS charges are not part of ActewAGL Distribution's regulated revenue requirement. Clause 6.18.7(a) of the transitional *Rules* allows ActewAGL Distribution to pass on to customers the charges to be incurred by ActewAGL Distribution for transmission use of system services. Clause 6.18.7(b) of the transitional *Rules* says that the amount to be passed on must not exceed the estimated amount of the TUOS charges for the relevant regulatory year adjusted for under or over recovery in the previous regulatory year. Clause 6.18.7(c) describes the method to be applied in determining the extent of under or over recovery.

To demonstrate compliance with clause 6.18.7 of the transitional *Rules* and the AER's final decision, the AER requires ActewAGL Distribution to maintain a transmission use of system (TUOS) overs and unders account. Clause 6.18.2(b)(7) requires ActewAGL Distribution to provide information on this account as part of the annual pricing proposal. The AER requires ActewAGL to provide details of its calculations in the format set out in Table E.1 of the final decision. Table 3.6 provides the required information.

Table 3.6 TUOS overs and unders account

	2009/10 Actual	2010/11 Estimate	2011/12 Forecast
Revenue from TUOS charges	29,693	36,440	42,430
Transmission charges to be paid to TNSPs	28,314	36,332	42,037
Avoided TUOS payments	0	0	0
Inter-DNSP payments	0	0	0
Total transmission related payments	28,314	36,332	42,037
Over (under) recovery for the financial year	1,380	108	393
Overs and unders account			
Annual rate of interest applicable to balances	8.79%	8.79%	8.79%
Semi-annual interest rate	4.30%	4.30%	4.30%
Opening Balance	-1,737	-450	-377
Interest on opening balance	-153	-40	-33
Over/under recovery for financial year	1,380	108	393
Interest on over/under recovery	59	5	17
Closing balance	-450	-377	0

The forecast revenue from TUOS charges for 2011/12 shown in Table 3.6 is \$42,430,016, an increase of 16.4 per cent compared to forecast TUOS revenue for 2010/11. This amount has been calculated on the basis of TransGrid's forecast prices for 2011/12 and the forecast energy imported into the ACT and includes the recovery of \$393,174 based on a forecast under-collection in 2010/11.

Based on TransGrid's prices, ActewAGL Distribution has forecast TUOS charges would be \$42,036,842 in 2011/12, an increase of 15.7 per cent. ActewAGL Distribution recovers TransGrid's TUOS charges from ACT consumers according to the energy they consume and, where possible, according to maximum demand in a month and over the year (where capacity charges apply). The cost allocations take into account the load profile of each customer class.

Only one set of TUOS prices is calculated, incorporating any adjustments to the total TUOS revenue requirement resulting from over or under recovery of TUOS charges in the previous regulatory year. The consumption profile used to calculate TUOS prices is the same 2010 consumption profile used to calculate DUOS prices.

As the energy transported in 2011/12 is forecast to be about 0.3 per cent higher¹⁹ than the actual energy transported in the 2010 calendar year, the amount of TUOS to be recovered in prices using the load in 2010 needs to be adjusted to \$42,319,304 or about 0.3 per cent lower.²⁰ This amount has been used to calculate the TUOS prices in Table 3.7. The TUOS prices would have recovered revenue of \$42,319,296 under the 2010 profile.

¹⁹ This is based on the forecasts adopted by the AER in the Final Distribution Determination – see chapter 7.

²⁰ AER, *Final decision, Control mechanisms for direct control services for the ACT and NSW 2009 distribution determinations*, February 2008, part 5.8.

Table 3.7 Transmission use of system charges 2011/12

<i>Description</i>	<i>Unit</i>	<i>2010 kWh/ cust no /kVA</i>	<i>Price 2011/12</i>	<i>Notional TUOS revenue</i>
Residential tariffs				
010 Residential Basic Network				
Network access charge	cents/day	130,120	0.00	\$0
Energy consumption	cents/kWh	926,040,359	1.377	\$12,751,576
015 Residential TOU Network				
Network access charge	cents/day	178	0.00	\$0
Energy at max times	cents/kWh	830,207	1.719	\$14,271
Energy at mid times	cents/kWh	953,105	1.376	\$13,115
Energy at economy times	cents/kWh	1,496,274	1.140	\$17,058
020 Residential 5000 Network				
Network access charge	cents/day	8,352	0.00	\$0
Energy for the first 60 kWh per day	cents/kWh	66,372,997	1.377	\$913,956
Energy above 60 kWh per day	cents/kWh	2,376,217	1.377	\$32,721
030 Residential with Heat Pump Network				
Network access charge	cents/day	4,573	0.00	\$0
Energy for the first 165 kWh per day	cents/kWh	82,003,467	1.377	\$1,129,188
Energy above 165 kWh per day	cents/kWh	1,207,635	1.377	\$16,629
060 Off-Peak (1) Night Network				
Energy consumption	cents/kWh	15,878,911	1.010	\$160,377
070 Off-Peak (3) Day & Night Network				
Energy consumption	cents/kWh	89,804,626	1.010	\$907,027
Renewable Energy Generation				
Energy consumption	cents/kWh	465,590	0.000	\$0
Commercial low voltage tariffs				
040 General Network				
Network access charge	cents/day	12,319	0.00	\$0
Energy for the first 330 kWh per day	cents/kWh	278,164,407	1.790	\$4,979,143
Energy above 330 kWh per day	cents/kWh	26,542,647	1.931	\$512,539
135 Small Unmetered Loads Network				
Network access charge	cents/day	23	0.00	\$0
Energy consumption	cents/kWh	1,352,100	1.549	\$20,944
141 Internal Network (Obsolete)				
Network access charge	cents/day	98	0.00	\$0
Energy consumption	cents/kWh	706,316	1.790	\$12,643
080 Streetlighting Network				
Network access charge	cents/day	18	0.00	\$0
Energy consumption	cents/kWh	40,151,345	1.343	\$539,233
090 General TOU Network				
Network access charge	cents/day	303	0.00	\$0
Energy at business times	cents/kWh	16,441,439	2.138	\$351,518
Energy at evening times	cents/kWh	5,915,492	1.379	\$81,575
Energy at off-peak times	cents/kWh	18,748,938	0.993	\$186,177
Low voltage time of use demand network				
101 LV TOU kVA Demand Network				
Network access per connection point	cents/day	1,610	0.00	\$0
Maximum demand charge	c/kVA/day	232,185	5.59	\$4,737,392
Energy at business times	cents/kWh	364,538,852	1.140	\$4,155,743
Energy at evening times	cents/kWh	125,965,345	1.095	\$1,379,321
Energy at off-peak times	cents/kWh	409,484,521	0.890	\$3,644,412
103 LV TOU Capacity Network				
Network access per connection point	cents/day	20	0.00	\$0
Maximum demand charge	c/kVA/day	6,520	2.57	\$61,195

<i>Description</i>	<i>Unit</i>	<i>2010 kWh/ cust no /kVA</i>	<i>Price 2011/12</i>	<i>Notional TUOS revenue</i>
Capacity charge	c/kVA/day	7,297	2.57	\$68,448
Energy at business times	cents/kWh	12,478,504	1.140	\$142,255
Energy at evening times	cents/kWh	5,729,845	1.095	\$62,742
Energy at off-peak times	cents/kWh	20,719,128	0.890	\$184,400
High voltage tariffs				
High voltage time of use demand network with ActewAGL low voltage network				
111 HV TOU Demand Network				
Network access per connection point	\$/day	1	0.00	\$0
Maximum demand charge	c/kVA/day	2,466	2.59	\$23,314
Capacity charge	c/kVA/day	3,039	2.59	\$28,732
Energy at business times	cents/kWh	4,223,276	1.089	\$45,991
Energy at evening times	cents/kWh	1,659,634	1.040	\$17,260
Energy at off-peak times	cents/kWh	5,338,577	0.841	\$44,897
112 HV TOU Demand Network – Customer HV				
Network access per connection point	\$/day	0	0.00	\$0
Maximum demand charge	c/kVA/day	0	2.59	\$0
Capacity charge	c/kVA/day	0	2.59	\$0
Energy at business times	cents/kWh	0	1.089	\$0
Energy at evening times	cents/kWh	0	1.040	\$0
Energy at off-peak times	cents/kWh	0	0.841	\$0
High voltage time of use demand network without ActewAGL low voltage network				
121 HV TOU Demand Network – Customer LV				
Network access per connection point	\$/day	21	0.00	\$0
Maximum demand charge	c/kVA/day	70,412	2.59	\$665,636
Capacity charge	c/kVA/day	86,838	2.59	\$820,924
Energy at business times	cents/kWh	138,441,490	1.089	\$1,507,628
Energy at evening times	cents/kWh	52,713,533	1.040	\$548,221
Energy at off-peak times	cents/kWh	183,249,415	0.841	\$1,541,128
122 HV TOU Demand Network – Customer HV and LV				
Network access per connection point	\$/day	0	0.00	\$0
Maximum demand charge	c/kVA/day	0	2.59	\$0
Capacity charge	c/kVA/day	0	2.59	\$0
Energy at business times	cents/kWh	0	1.089	\$0
Energy at evening times	cents/kWh	0	1.040	\$0
Energy at off-peak times	cents/kWh	0	0.841	\$0
Total				\$42,319,296

3.5 Network use of system charges

Network use of system (NUOS) charges for 2011/12 comprise the distribution use of system (DUOS) charges and transmission use of system (TUOS) charges. The NUOS charges are shown in Table 3.8. All charges exclude GST.

Table 3.8 Network use of system charges 2011/12

Code	Description	Unit	Distribution Charges 2011/12	Transmission Charges 2011/12	Network Charges 2011/12
Residential tariffs					
010	Residential Basic Network				
	Network access charge	cents/day	15.25	0.00	15.25
	Energy consumption	cents/kWh	4.933	1.377	6.310
015	Residential TOU Network				
	Network access charge	cents/day	15.25	0.00	15.25
	Energy consumption at max times	cents/kWh	7.461	1.719	9.180
	Energy consumption at mid times	cents/kWh	3.694	1.376	5.070
	Energy consumption at economy times	cents/kWh	2.600	1.140	3.740
020	Residential 5000 Network				
	Network access charge	cents/day	36.75	0.00	36.75
	Energy consumption for the first 60 kWh per day	cents/kWh	3.353	1.377	4.730
	Energy consumption above 60 kWh per day	cents/kWh	4.933	1.377	6.310
030	Residential with Heat Pump Network				
	Network access charge	cents/day	79.95	0.00	79.95
	Energy consumption for the first 165 kWh per day	cents/kWh	1.863	1.377	3.240
	Energy consumption above 165 kWh per day	cents/kWh	4.933	1.377	6.310
060	Off-Peak (1) Night Network				
	Energy consumption	cents/kWh	0.350	1.010	1.360
070	Off-Peak (3) Day & Night Network				
	Energy consumption	cents/kWh	0.830	1.010	1.840
	Renewable Energy Generation				
	Energy consumption	cents/kWh	0.000	0.000	0.000
Commercial low voltage tariffs					
040	General Network				
	Network access charge	cents/day	30.71	0.00	30.71
	Energy consumption for the first 330 kWh per day	cents/kWh	7.850	1.790	9.640
	Energy consumption above 330 kWh per day	cents/kWh	10.749	1.931	12.680
135	Small Unmetered Loads Network				
	Network access charge	cents/day	24.80	0.00	24.80
	Energy consumption	cents/kWh	9.331	1.549	10.880
141	Internal Network (Obsolete)				
	Network access charge	cents/day	30.71	0.00	30.71
	Energy consumption	cents/kWh	7.850	1.790	9.640
080	Streetlighting Network				
	Network access charge	cents/day	30.00	0.00	30.00
	Energy consumption	cents/kWh	5.097	1.343	6.440
090	General TOU Network				
	Network access charge	cents/day	30.71	0.00	30.71
	Energy consumption at business times	cents/kWh	13.152	2.138	15.290
	Energy consumption at evening times	cents/kWh	6.371	1.379	7.750
	Energy consumption at off-peak times	cents/kWh	2.337	0.993	3.330
Low voltage time of use demand network					
101	LV TOU kVA Demand Network				
	Network access charge per connection point	cents/day	36.00	0.00	36.00
	Maximum demand charge	c/kVA/day	34.81	5.59	40.40
	Energy consumption at business times	cents/kWh	3.660	1.140	4.800
	Energy consumption at evening times	cents/kWh	2.195	1.095	3.290

Code	Description	Unit	Distribution Charges 2011/12	Transmission Charges 2011/12	Network Charges 2011/12
103	Energy consumption at off-peak times	cents/kWh	0.800	0.890	1.690
	LV TOU Capacity Network				
	Network access charge per connection point	cents/day	36.00	0.00	36.00
	Maximum demand charge	c/kVA/day	16.03	2.57	18.60
	Capacity charge	c/kVA/day	16.03	2.57	18.60
	Energy consumption at business times	cents/kWh	3.660	1.140	4.800
	Energy consumption at evening times	cents/kWh	2.195	1.095	3.290
	Energy consumption at off-peak times	cents/kWh	0.800	0.890	1.690
High voltage tariffs					
High voltage time of use demand network with ActewAGL low voltage network					
111	HV TOU Demand Network				
	Network access charge per connection point	\$/day	13.00	0.00	13.00
	Maximum demand charge	c/kVA/day	13.91	2.59	16.50
	Capacity charge	c/kVA/day	13.91	2.59	16.50
	Energy consumption at business times	cents/kWh	3.071	1.089	4.160
	Energy consumption at evening times	cents/kWh	1.780	1.040	2.820
	Energy consumption at off-peak times	cents/kWh	0.649	0.841	1.490
112	HV TOU Demand Network – Customer HV				
	Network access charge per connection point	\$/day	13.00	0.00	13.00
	Maximum demand charge	c/kVA/day	12.91	2.59	15.50
	Capacity charge	c/kVA/day	12.91	2.59	15.50
	Energy consumption at business times	cents/kWh	3.071	1.089	4.160
	Energy consumption at evening times	cents/kWh	1.780	1.040	2.820
	Energy consumption at off-peak times	cents/kWh	0.649	0.841	1.490
High voltage time of use demand network without ActewAGL low voltage network					
121	HV TOU Demand Network – Customer LV				
	Network access charge per connection point	\$/day	13.00	0.00	13.00
	Maximum demand charge	c/kVA/day	13.91	2.59	16.50
	Capacity charge	c/kVA/day	13.91	2.59	16.50
	Energy consumption at business times	cents/kWh	2.661	1.089	3.750
	Energy consumption at evening times	cents/kWh	1.420	1.040	2.460
	Energy consumption at off-peak times	cents/kWh	0.499	0.841	1.340
122	HV TOU Demand Network – Customer HV and LV				
	Network access charge per connection point	\$/day	13.00	0.00	13.00
	Maximum demand charge	c/kVA/day	12.91	2.59	15.50
	Capacity charge	c/kVA/day	12.91	2.59	15.50
	Energy consumption at business times	cents/kWh	2.661	1.089	3.750
	Energy consumption at evening times	cents/kWh	1.420	1.040	2.460
	Energy consumption at off-peak times	cents/kWh	0.499	0.841	1.340

3.6 Changes to network tariffs

The transitional *Rules* (6.18.2(b)(8)) require an explanation of the nature and extent of changes from the previous regulatory year.

The changes from 2010/11 to the residential network tariff structure are as follows:

- ActewAGL Distribution has made time-of-use network tariffs the default tariffs for new premises. ActewAGL Distribution supports the application of time-of-use tariffs to new

customers moving into premises with time-of-use meters installed. It also encourages retailers to review the bills of customers with time-of-use meters installed with a view to shifting them to time-of-use tariffs when the customer would be better off.

- These movements result in a minor loss of revenue for ActewAGL for the current financial year.
- ActewAGL Distribution is also extending eligibility for the *Residential TOU* tariff to include electric vehicle recharging facilities in residential premises by amending its pricing schedule. This change comes in response to the decision by Better Place Australia to choose Canberra as the site of Better Place Australia's first city-wide roll-out of electric vehicle infrastructure in Australia, the availability of electric vehicles, and increasing public awareness of electric vehicles as an alternative to conventional internal combustion engine vehicles. It is intended that this change will allow customers with electric vehicles to access the benefits of a TOU network tariff, whilst also assisting ActewAGL in managing the demands on its electricity network.

Table 3.9 compares the network charges in 2011/12 with those in 2010/11. The average change in network charges is shown in cents per kWh and as a percentage for an average consumer for each tariff.²¹

²¹ The average change in network charges is calculated by determining the average revenue for each tariff using the prices for each year and taking the difference. The percentage change is this difference divided by the average revenue with 2008/09 prices. Where there are no customers, it is not possible to calculate the average price.

Table 3.9 Changes in network charges

<i>Cod</i>	<i>Description</i>	<i>Unit</i>	<i>Network Charges 2010/11</i>	<i>Network Charges 2011/12</i>	<i>Average change c/kWh</i>	<i>Average change %</i>
Residential tariffs						
010	Residential Basic Network				0.524	8.0%
	Network access charge	cents/day	14.01	15.25		
	Energy consumption	cents/kWh	5.85	6.31		
015	Residential TOU Network				0.403	7.5%
	Network access charge	cents/day	14.01	15.25		
	Energy consumption at max times	cents/kWh	8.43	9.18		
	Energy consumption at mid times	cents/kWh	4.75	5.07		
	Energy consumption at economy times	cents/kWh	3.53	3.74		
020	Residential 5000 Network				0.534	9.1%
	Network access charge	cents/day	35.51	36.75		
	Energy consumption for the first 60 kWh per day	cents/kWh	4.25	4.73		
	Energy consumption above 60 kWh per day	cents/kWh	5.85	6.31		
030	Residential with Heat Pump Network				0.528	12.1%
	Network access charge	cents/day	78.51	79.95		
	Energy consumption for the first 165 kWh per day	cents/kWh	2.74	3.24		
	Energy consumption above 165 kWh per day	cents/kWh	5.85	6.31		
060	Off-Peak (1) Night Network				0.200	17.2%
	Energy consumption	cents/kWh	1.16	1.36		
070	Off-Peak (3) Day & Night Network				0.280	17.9%
	Energy consumption	cents/kWh	1.56	1.84		
	Renewable Energy Generation					
	Energy consumption	cents/kWh	0.00	0.00		
Commercial low voltage						
040	General Network				0.67	6.9%
	Network access charge	cents/day	27.66	30.71		
	Energy consumption for the first 330 kWh per day	cents/kWh	9.03	9.64		
	Energy consumption above 330 kWh per day	cents/kWh	11.90	12.68		
135	Small Unmetered Loads Network				1.118	11.3%
	Network access charge	cents/day	20.30	24.80		
	Energy consumption	cents/kWh	9.79	10.88		
141	Internal Network (Obsolete)				0.765	7.3%
	Network access charge	cents/day	27.66	30.71		
	Energy consumption	cents/kWh	9.03	9.64		
080	Streetlighting Network				1.070	19.9%
	Network access charge	cents/day	28.00	30.00		
	Energy consumption	cents/kWh	5.37	6.44		
090	General TOU Network				0.700	8.6%
	Network access charge	cents/day	27.66	30.71		
	Energy consumption at business times	cents/kWh	14.00	15.29		
	Energy consumption at evening times	cents/kWh	7.32	7.75		
	Energy consumption at off-peak times	cents/kWh	3.08	3.33		
Low voltage time of use demand network						
101	LV TOU kVA Demand Network				0.486	7.5%
	Network access charge per connection point	cents/day	35.00	36.00		
	Maximum demand charge	c/kVA/day	37.60	40.40		

<i>Codi</i>	<i>Description</i>	<i>Unit</i>	<i>Network Charges 2010/11</i>	<i>Network Charges 2011/12</i>	<i>Average change c/kWh</i>	<i>Average change %</i>
103	Energy consumption at business times	cents/kWh	4.42	4.80	0.364	7.3%
	Energy consumption at evening times	cents/kWh	2.97	3.29		
	Energy consumption at off-peak times	cents/kWh	1.64	1.69		
	LV TOU Capacity Network					
	Network access charge per connection point	cents/day	35.00	36.00		
	Maximum demand charge	c/kVA/day	17.30	18.60		
	Capacity charge	c/kVA/day	17.30	18.60		
	Energy consumption at business times	cents/kWh	4.42	4.80		
	Energy consumption at evening times	cents/kWh	2.97	3.29		
	Energy consumption at off-peak times	cents/kWh	1.64	1.69		
High voltage tariffs						
High voltage time of use demand network with ActewAGL low voltage network						
111	HV TOU Demand Network				0.335	6.3%
	Network access charge per connection point	\$/day	13.00	13.00		
	Maximum demand charge	c/kVA/day	15.20	16.50		
	Capacity charge	c/kVA/day	15.20	16.50		
	Energy consumption at business times	cents/kWh	3.95	4.16		
	Energy consumption at evening times	cents/kWh	2.63	2.82		
	Energy consumption at off-peak times	cents/kWh	1.50	1.49		
112	HV TOU Demand Network – Customer HV					
	Network access charge per connection point	\$/day	13.00	13.00		
	Maximum demand charge	c/kVA/day	14.20	15.50		
	Capacity charge	c/kVA/day	14.20	15.50		
	Energy consumption at business times	cents/kWh	3.95	4.16		
	Energy consumption at evening times	cents/kWh	2.63	2.82		
	Energy consumption at off-peak times	cents/kWh	1.50	1.49		
High voltage time of use demand network without ActewAGL low voltage network						
121	HV TOU Demand Network – Customer LV				0.309	6.70%
	Network access charge per connection point	\$/day	13.00	13.00		
	Maximum demand charge	c/kVA/day	15.20	16.50		
	Capacity charge	c/kVA/day	15.20	16.50		
	Energy consumption at business times	cents/kWh	3.53	3.75		
	Energy consumption at evening times	cents/kWh	2.26	2.46		
	Energy consumption at off-peak times	cents/kWh	1.34	1.34		
122	HV TOU Demand Network – Customer HV and LV					
	Network access charge per connection point	\$/day	13.00	13.00		
	Maximum demand charge	c/kVA/day	14.20	15.50		
	Capacity charge	c/kVA/day	14.20	15.50		
	Energy consumption at business times	cents/kWh	3.53	3.75		
	Energy consumption at evening times	cents/kWh	2.26	2.46		
	Energy consumption at off-peak times	cents/kWh	1.34	1.34		

The increases shown in the Table 3.9 reflect the increases in both DUOS and TUOS charges for 2011/12. The CPI of 2.85 per cent and the X factor of -4.00 per cent raise the MAAR by 6.96 per cent. In addition, there is a 16.6 per cent increase in average TUOS charges in 2011/12. The negative pass through of the excess requirements of funds to meet the cost of the ACT feed-in tariff has lowered the average increase in NUOS charges by 0.075 c/kWh or 1.4 per cent.

Table 3.10 compares the miscellaneous fees and charges for 2011/12 with the comparable charges for 2010/11. The average increase in these miscellaneous fees and charges is around 3.1 per cent.

Table 3.10 Changes in miscellaneous services charges

Code	Description	Unit	Prices 2010/11	Prices 2011/12	Change %
For visit to re-energise or de-energise a site					
501	Business hours – re-energise a site	per visit	\$49.09	\$50.91	3.7%
502	After hours – re-energise a site	per visit	\$105.45	\$109.09	3.5%
503	Business hours – de-energise a site	per visit	\$43.64	\$44.55	2.1%
505	De-energise a site for non-payment	per visit	\$81.82	\$84.55	3.3%
508	Field visit only (De-energise a site for non-payment)	per visit	\$59.09	\$60.91	3.1%
Temporary connections					
520	Overhead	per installation	\$350.00	\$360.91	3.1%
521	Standard underground	per installation	\$508.18	\$523.64	3.0%
522	Free-standing underground	per installation	\$616.36	\$635.45	3.1%
Services					
530	Overhead: remove, reposition or disconnect service	per site visit	\$251.82	\$260.00	3.2%
531	Underground: remove, reposition or disconnect service	per site visit	\$606.36	\$625.45	3.1%
Upgrade service from single to three phase at customer's request where load does not justify three phase					
532	Overhead	per installation	\$325.45	\$335.45	3.1%
533	Underground – service cable replacement not required	per installation	\$325.45	\$335.45	3.1%
534	Underground – service cable replacement required	per installation	\$606.36	\$625.45	3.1%
540	Rescheduled visit (applied where a revisit to a site is necessitated by obstructed access, non-compliance with the Service and Installation Rules or the client is not ready for the scheduled work.)				
550	Issue of copies of electrical drawings	per sheet or electronic equivalent	\$167.27	\$172.73	3.3%
560	De-energising wires (to allow safe approach, for example, for tree pruning, plant operation, oversize loads, construction activities)	per occurrence	\$404.55	\$417.27	3.1%
570	Embedded generation - operational and maintenance charges for small embedded generators (other renewable <30KW)				
	for connection assets		2.0%	2.0%	0.0%
	for shared network assets (if applicable)		2.0%	2.0%	0.0%

3.6.1 Proposed Guarantee of Minimum Revenue scheme

To ensure continuing efficiency of network investments and to better protect its network investment and the wider ACT electricity distribution customer base, ActewAGL Distribution is in the initial stages of developing a Guarantee of Minimum Revenue (GoR) scheme. The scheme's main aims are to manage ActewAGL's network investment risk, by both reducing the risk of stranded assets and reducing the risk to all customers of carrying the cost of specific higher risk projects. It is intended that these objectives will be achieved by providing a pricing (incentive) signal at the project development stage to achieve a more efficient investment outcome than might otherwise be achieved.

ActewAGL analysis of the experience of distributors in other jurisdictions suggests that schemes such as the GoR can be successful in achieving these aims.

To date, ActewAGL has undertaken preliminary assessments of the impacts of the proposed scheme. The findings of these assessments are:

- The scheme is designed to limit the cost impact on the customer whilst still promoting responsible behaviour in respect of the customer's demand estimates; and
- The scheme is not a revenue raising measure and would therefore not have implications for compliance with the MAAR .

4. Charges for metering services

4.1 The structure and basis of ActewAGL Distribution's metering charges

ActewAGL Distribution's schedule of metering charges comprises daily charges for the provision of each of four different types of manually read metering services and charges for special meter reads. Meters for residential consumers are usually read quarterly while those for small low voltage commercial consumers (consuming less than 160 MWh pa) are usually read monthly. Metering services for larger low voltage consumers and for high voltage consumers are open to competition and are not regulated.

The metering charges for small consumers are set on a cost reflective basis. The costs of meter reading and meters (including depreciation, return on capital, installation and maintenance and administration) have been calculated for each meter type. The costs are applied to the number of meters of each type to derive the total cost for each meter type. The meter types are each allocated to a class of metering service and the total cost of each class is allocated to the customers in that class to estimate the average cost of the metering service to each class.

4.2 Metering services charges for 2011/12

Prices for ActewAGL Distribution's alternative control services are regulated using a maximum revenue cap.²² This cap is applied to metering revenue that would have been earned from the prices for the coming financial year if they had been applied to the quantities sold in the services provided in the previous calendar year.

Table 4.1 shows the X factor for 2011/12 and CPI factors applied to calculate the revenue cap for alternative control services. The approach is similar to that applied in Table 3.1 to calculate the MAAR.

Table 4.1 Alternative control services revenue cap

	<i>Alternative services revenue in previous year</i>	<i>X Factor</i>	<i>Sum of CPI indices</i>	<i>CPI</i>	<i>Alternative service revenue cap</i>
2008/09			631.8		\$5,666,539
2009/10	\$5,666,539	-29.30%	659.3	4.35%	\$7,645,746
2010/11	\$7,645,746	0.00%	671.3	1.82%	\$7,784,907
2011/12	\$7,784,907	0.00%	690.4	2.85%	\$8,006,405

²² AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 146

As shown in Table 4.1, the alternative control services revenue cap for 2011/12 is \$8,342,674.

To demonstrate compliance with this revenue cap, ActewAGL Distribution has prepared the required schedule of metering charges showing the revenue from each charge if that charge had applied in the 2010 calendar year.²³ The schedule shows that the revenue from these charges does not exceed the revenue cap for alternative control services. The metering charges and the forecast revenue from those charges are shown in Table 4.2.

Table 4.2 Metering charges 2011/12

Code	Description	Unit	Units 2010	Prices excl. GST 2011/12	Revenue
MP1	Quarterly basic metering rate Accumulation and time-of-use meters read quarterly	cents per day per NMI	143,206	12.75	\$6,664,450
MP2	Monthly basic metering rate Accumulation and time-of-use meters read monthly	cents per day per NMI	12,093	22.29	\$983,851
MP3	Time-of-use metering rate Time-of-use meters read monthly	cents per day per NMI	116	22.29	\$9,431
MP4	Monthly manually-read interval metering rate Interval meters recording at either 15 or 30 minute intervals, read manually and processed monthly	\$ per day per NMI	252	\$1.81	\$166,359
MP5	Internal metering rate Sites entitled to the Internal Network charge	cents per day per NMI	97	22.29	\$7,884
MP6	Quarterly manually-read interval metering rate Interval meters recording at either 15 or 30 minute intervals, read manually and processed quarterly	cents per day per NMI	0	52.00	\$0
504	Meter test during business hours (deposit)	per test	31	\$61.82	\$1,928
506	Special meter reading, for example, check read (second in a billing period)	per read	0	\$31.82	\$0
507	Install interval meter at customer's request	per meter	0	\$153.64	\$0
509	Replace meter to facilitate renewable energy installation	per meter	1,122	\$153.64	\$172,429
Total			156,917		\$8,006,331

*National Meter Identifier.

The revenue from the 2010 units of \$8,006,331 is less than the revenue cap for alternative control services of \$8,006,405. Therefore, the charges comply with the AER's Final Distribution Determination on alternative control services.

4.3 Changes in metering charges

There are no changes to metering services or to the structure of metering charges in 2011/12.

²³ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 153.

Table 4.3 compares the metering charges for 2011/12 with those for 2010/11.

Table 4.3 Changes in metering charges

Code	Description	Unit	Prices 2010/11	Prices 2011/12	Change %
MP1	Quarterly basic metering rate				
	Accumulation and time-of-use meters read quarterly	cents per day per NMI	12.79	12.75	-0.3%
MP2	Monthly basic metering rate				
	Accumulation and time-of-use meters read monthly	cents per day per NMI	22.34	22.29	-0.2%
MP3	Time-of-use metering rate				
	Time-of-use meters read monthly	cents per day per NMI	22.34	22.29	-0.2%
MP4	Monthly manually-read interval metering rate				
	Interval meters recording at either 15- or 30-minute intervals, read manually and processed monthly	\$ per day per NMI	\$1.82	\$1.81	-0.5%
MP5	Internal metering rate				
	Sites entitled to the Internal Network charge	cents per day per NMI	22.34	22.29	-0.2%
MP6	Quarterly manually-read interval metering rate				
	Interval meters recording at either 15- or 30-minute intervals, read manually and processed quarterly	cents per day per NMI	52.50	52.00	-1.0%
504	Meter test during business hours (deposit)	Per test	\$60.91	\$61.82	1.5%
506	Special meter reading , for example, check read (second in a billing period)	per read	\$31.82	\$31.82	0.0%
507	Install interval meter at customer's request	per meter	\$152.73	\$153.64	0.6%
509	Replace meter to facilitate renewable energy installation	per meter	\$152.73	\$153.64	0.6%

5. Indicative customer impacts

5.1 Changes in network and metering charges

ActewAGL Distribution's customers who consume less than 160 MWh per annum are most likely to be subject to charges for both standard control services (network tariffs) and alternative control services (regulated metering service charges). Table 5.1 shows network charges (standard control services and TUOS) plus metering service charges (alternative control services) for 2011/12 and the comparable charges for 2010/11, excluding GST. High voltage charges have been excluded as ActewAGL Distribution does not provide metering services to high voltage customers. Metering services to customers consuming more than 160 MWh per annum are open to competition and not regulated.

Standard control services and alternative control services are shown separately in ActewAGL Distribution's pricing schedule and are charged separately to retailers.

Table 5.1 Network and metering charges 2011/12

Code	Description	Unit	Network & metering charges 2010/11	Network & metering charges 2011/12	Average change c/kWh	Average change %
Residential tariffs						
010	Residential Basic Network				0.52	7.2%
	Network access charge	cents/day	26.80	28.00		
	Energy consumption	cents/kWh	5.85	6.31		
015	Residential TOU Network				0.40	7.1%
	Network access charge	cents/day	26.80	28.00		
	Energy consumption at max times	cents/kWh	8.43	9.18		
	Energy consumption at mid times	cents/kWh	4.75	5.07		
	Energy consumption at economy times	cents/kWh	3.53	3.74		
020	Residential 5000 Network				0.53	8.3%
	Network access charge	cents/day	48.30	49.50		
	Energy consumption for the first 60 kWh per day	cents/kWh	4.25	4.73		
	Energy consumption above 60 kWh per day	cents/kWh	5.85	6.31		
030	Residential with Heat Pump Network				0.53	11.4%
	Network access charge	cents/day	91.30	92.70		
	Energy consumption for the first 165 kWh per day	cents/kWh	2.74	3.24		
	Energy consumption above 165 kWh per day	cents/kWh	5.85	6.31		
060	Off-Peak (1) Night Network				0.20	17.2%
	Energy consumption	cents/kWh	1.16	1.36		
070	Off-Peak (3) Day & Night Network				0.28	17.9%
	Energy consumption	cents/kWh	1.56	1.84		
	Renewable Energy Generation					
	Energy consumption	cents/kWh	0.00	0.00		

Code	Description	Unit	Network & metering charges 2010/11	Network & metering charges 2011/12	Average change c/kWh	Average change %
Commercial low voltage tariffs						
040	General Network				0.67	6.7%
	Network access charge	cents/day	50.00	53.00		
	Energy consumption for the first 330 kWh per day	cents/kWh	9.03	9.64		
	Energy consumption above 330 kWh per day	cents/kWh	11.90	12.68		
135	Small Unmetered Loads Network				1.12	11.3%
	Network access charge	cents/day	20.30	24.80		
	Energy consumption	cents/kWh	9.79	10.88		
080	Streetlighting Network				1.07	19.9%
	Network access charge	cents/day	50.34	52.29		
	Energy consumption	cents/kWh	5.37	6.44		
090	General TOU Network				0.70	8.5%
	Network access charge	cents/day	50.00	53.00		
	Energy consumption at business times	cents/kWh	14.00	15.29		
	Energy consumption at evening times	cents/kWh	7.32	7.75		
	Energy consumption at off-peak times	cents/kWh	3.08	3.33		
Low voltage time of use demand network						
101	LV TOU kVA Demand Network				0.49	7.3%
	Network access charge per connection point	cents/day	217	217		
	Maximum demand charge	c/kVA/day	37.60	40.40		
	Energy consumption at business times	cents/kWh	4.42	4.80		
	Energy consumption at evening times	cents/kWh	2.97	3.29		
	Energy consumption at off-peak times	cents/kWh	1.64	1.69		
103	LV TOU Capacity Network				0.36	7.3%
	Network access charge per connection point	cents/day	217	217		
	Maximum demand charge	c/kVA/day	17.30	18.60		
	Capacity charge	c/kVA/day	17.30	18.60		
	Energy consumption at business times	cents/kWh	4.42	4.80		
	Energy consumption at evening times	cents/kWh	2.97	3.29		
	Energy consumption at off-peak times	cents/kWh	1.64	1.69		

5.2 Estimated impacts on average customer electricity bills

The 2011/12 increases in network and metering charges would raise the electricity bill for an average residential customer consuming 7000 kWh on the Residential Basic network charge by \$0.77 per week (including GST), a real increase of 4.2 per cent (7.2 per cent nominal). For a commercial customer consuming 30 MWh per annum on General network charge, the network and metering price increases would raise their electricity bill by \$4.10 per week (including GST) or 3.8 per cent real (6.7 per cent nominal).

Attachment 1: Compliance with transitional Rules requirements

Table A1.1 provides a checklist of where the relevant requirements of Part I of the transitional *Rules* are met in this document. One of the transitional *Rules* requirements is that the pricing proposal demonstrates compliance with any applicable distribution determination (clause 6.18.2(7)). Table A1.2 provides a separate checklist of the relevant requirements from the AER's *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision* and where they are addressed in this document.

Table A1.1: Checklist of transitional Rules pricing proposal requirements

Rules	Requirement	Coverage in this document
6.18.2 (b)	The pricing proposal must: (1) set out the tariff classes that are to apply for the relevant regulatory year; and (2) set out the proposed tariffs for each tariff class; and (3) set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates; and (4) set out, for each tariff class related to standard control services, the expected weighted average revenue for the relevant regulatory year and also for the current regulatory year; and (5) set out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur; and (6) set out how charges incurred by the Distribution Network Service Provider for transmission use of system services are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year; and (7) demonstrate compliance with the Rules and any applicable distribution determination; and (8) describe the nature and extent of change from the previous regulatory year and demonstrate that the changes comply with the Rules and any applicable distribution determination.	(1) The tariff classes for standard control services are set out in Section 2.1. (2) Distribution tariffs are in Table 3.3. Metering charges are in Table 4.2. (3) Section 2.1 and Tables 2.1, 2.2 and 2.3 set out each charging parameter and the element of service to which it relates. (4) Table 3.4 shows the weighted average DOUS revenue for 2010/11 and 2011/12. (5) In the event that the Federal Court's decision in relation to the pending application (noted in section 1.3) is to set aside the averaging period commencing 2 February 2009, the tariffs will need to be recalculated on the basis of a nominal vanilla WACC of 10.05 per cent (or other WACC found applicable by the Court) and adjusted accordingly. ActewAGL Distribution does not anticipate any further variations or adjustments to tariffs for 2011/12. (6) Section 3.4 provides an explanation of how TUOS charges are passed on to customers, and ActewAGL Distribution's adjustment for under recovery of TUOS costs in 2011/12. TUOS charges are provided in Table 3.7 (7) Table A1.2 below provides the checklist for relevant requirements from the 2009-14 distribution determination. (8) Sections 3.6 and 4.3 contain descriptions of the nature and extent of changes from the 2010/11 regulatory year. Table 3.3 and Table 3.5 demonstrate that revenue from 2011/12 prices matches allowable revenue calculated in section 3.1.1
6.18.3	(a) A pricing proposal must define the tariff classes into which customers for direct control services are divided. (b) Each customer for direct control services must be a member of 1 or more tariff classes. (c) Separate tariff classes must be constituted for customers to whom standard control services are supplied and customers to whom alternative control services are supplied (but a customer for both standard control services and alternative control services may be a member of 2 or more tariff classes). (d) A tariff class must be constituted with regard to: (i) the need to group customers together on an economically efficient basis; and (ii) the need to avoid unnecessary	(a) Section 2.1 provides an explanation of the tariff classes for standard control services. Section 4.1 provides an explanation of the categories of charges for alternative control services. (b) Each customer is on one or more tariffs within one or more tariff classes for which qualify (c) Separate tariff classes and charges are specified for standard control services in Table 3.3 and Table 3.5 and alternative control services in Table 4.2. (d) Section 2.1 contains an explanation of the basis of the tariff classes.

	transaction costs.	
6.18.4	(a) In formulating provisions of a distribution determination governing the assignment of customers to tariff classes or the re-assignment of customers from one tariff class to another, the AER must have regard to the following principles: (1) customers should be assigned to tariff classes on the basis of one or more of the following factors: (i) the nature and extent of their usage; (ii) the nature of their connection to the network; (iii) whether remotely-read interval metering or other similar metering technology has been installed at the customer's premises as a result of a regulatory obligation or requirement; (2) customers with a similar connection and usage profile should be treated on an equal basis; (3) however, customers with micro-generation facilities should be treated no less favourably than customers without such facilities but with a similar load profile; (4) a Distribution Network Service Provider's decision to assign a customer to a particular tariff class, or to re-assign a customer from one tariff class to another should be subject to an effective system of assessment and review.	(a) Appendix A of the AER's final decision sets out the procedures for assigning or re-assigning customers to tariff classes.
6.18.5	(a) For each tariff class, the revenue expected to be recovered should lie on or between: (1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and (2) a lower bound representing the avoidable cost of not serving those customers. (b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class: (1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter relates; and (2) must be determined having regard to: (i) transaction costs associated with the tariff or each charging parameter; and (ii) whether customers of the relevant tariff class are able or likely to respond to price signals. (c) If, however, as a result of the operation of paragraph (b), the Distribution Network Service Provider may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.	(a) Table 2.4 shows that the revenue expected from each tariff class lies between the stand alone cost and the avoidable cost for the tariff class. Section 2.3 contains an explanation of the basis and application of the principle. (b) Section 2.3 contains explanations of how ActewAGL Distribution takes into account long run marginal costs, transactions costs, the scope for customers to respond to price signals and the need to recover revenue in an efficient manner (as required by (c) below). (c) Section 2.3 contains explanations of how the need to recover revenue with minimum distortion to efficient patterns of consumption is taken into account when setting charging parameters.
6.18.6	(a) This clause applies only to <i>tariff classes</i> related to	Section 3.2 and Table 3.4 show that tariff classes comply with

	the provision of <i>standard control services</i>. (b) The expected weighted average revenue to be raised from a <i>tariff class</i> for a particular <i>regulatory year</i> of a <i>regulatory control period</i> must not exceed the corresponding expected weighted average revenue for the preceding <i>regulatory year</i> by more than the permissible percentage. (c) The permissible percentage is the greater of the following: (1) the CPI-X limitation on any increase in the <i>Distribution Network Service Provider's</i> expected weighted average revenue between the two <i>regulatory years</i> plus 2%; Note: The calculation is of the form $(1 + \text{CPI})(1 - X)(1 + 2\%)$ (2) CPI plus 2%. Note: The calculation is of the form $(1 + \text{CPI})(1 + 2\%)$ (d) In deciding whether the permissible percentage has been exceeded in a particular regulatory year, the following are to be disregarded: (1) the recovery of revenue to accommodate a variation to the distribution determination under rule 6.6 or 6.13; (2) the recovery of revenue to accommodate pass through of charges for transmission use of system services to customers. (e) This clause does not, however, limit the extent a tariff for customers with remotely-read interval metering or other similar metering technology may vary according to the time or other circumstances of the customer's usage.	the side constraints.
6.18.7	(a) A <i>pricing proposal</i> must provide for tariffs designed to pass on to customers the charges to be incurred by the <i>Distribution Network Service Provider</i> for <i>transmission use of system services</i>. (b) The amount to be passed on to customers for a particular <i>regulatory year</i> must not exceed the estimated amount of the <i>transmission use of system charges</i> for the relevant <i>regulatory year</i> adjusted for over or under recovery in the previous <i>regulatory year</i>. (c) The extent of the over or under recovery is the difference between: (1) the amount actually paid by the <i>Distribution Network Service Provider</i> by way of <i>transmission use of system charges</i> in the previous <i>regulatory year</i>; and (2) the amount passed on to customers by way of <i>transmission use of system charges</i> by the <i>Distribution Network Service Provider</i> in the previous <i>regulatory year</i>.	Section 3.4 provides an explanation of how TUOS charges are passed on to customers, and ActewAGL Distribution's adjustment for under recovery of TUOS costs in 2010/11. TUOS charges are provided in Table 3.7.

Table A1.2: Checklist of requirements from the AER final decision

<i>Final decision requirement</i>	<i>Coverage in this document</i>
<i>Compliance with X factors</i> "In accordance with clause 6.12.1(11) of the transitional chapter 6 rules the X factors to apply to ActewAGL are set out in table 17.8 of the final decision" (AER Final Decision, p. 145)	Table 3.1 shows that the X factor applied to the calculation of MAAR for standard control services is - 4.00% and therefore complies with the AER's final decision. Section 4.2 demonstrates that an X factor of 0.0% has been applied in calculating the revenue cap for alternative control services. This complies with the AER's Final Decision.
<i>Compliance with the standard control services control mechanism.</i> "In accordance with clause 6.12.1(13) of the transitional chapter 6 rules compliance with the maximum allowable average revenue cap for standard control services must be demonstrated by ActewAGL using the formulae outlined in section 4.5 and in accordance with appendix F of this final decision." (AER Final Decision, p. 20)	Section 3.1, shows ActewAGL Distribution's application of the formulae set out in section 4.5 of the AER final decision to derive the revenue cap for standard control services. Section 3.2 and Table 3.3 demonstrate that the revenue from standard control services derived from 2011/12 tariffs complies with this cap. No adjustments under Appendix F have been made. Time-of-use prices have been made the default tariff for new premises and this data will feed into the pricing load profile over time.
<i>Compliance with the alternative control services control mechanism.</i> "In accordance with clause 6.12.1(13) of the transitional chapter 6 rules ActewAGL must demonstrate compliance with the control mechanism for alternative control services by submitting to the AER a schedule of metering charges, in the form of table 13.5 of ActewAGL's revenue proposal, as soon as practicable after prices for each regulatory year are determined." (AER Final Decision, p. 123)	Section 4.2 contains Table 4.2 which is in the same format as Table 13.5 in ActewAGL Distribution's regulatory proposal.
<i>TUOS unders and overs account.</i> "In accordance with clause 6.12.1(19) of the transitional chapter 6 rules ActewAGL must submit, as part of its annual pricing proposal, a record of the amount of revenues recovered from TUOS charges and associated payments in accordance with appendix E of this final decision." (AER Final Decision, p. 20)	The TUOS unders and overs account is shown in section 3.4 (Table 3.6).
<i>Side constraints</i> "In accordance with clause 6.18.2(b)(7) and (8) of the transitional chapter 6 rules ActewAGL must ensure its annual pricing proposals comply with the side constraint formula defined in section 4.5 of this final decision." (AER final decision, p. 20)	Side constraints are considered in section 3.2 and compliance is shown in Table 3.4.